

Reading the world on intraday MNQ and MGC

GAZBOT V7 · night of Saturday 2026-09-13 · a **reading** report, not a backtest. Fifteen research lenses plus three syntheses, ~340 catalogued claims, ~290 distinct sources, 13 academic papers read in full text, ~155 forum pages, every prop firm's own rulebook, and every publicly-verified intraday futures track record I could reach. Costs charged throughout: **MNQ 1.25 points all-in per round trip** (\$1.50 commission + the 0.50pt spread crossed once = \$2.50). **MGC 0.45 points = \$4.50**. Notes for every lens live in `reports/lit_review_2026-09-13/<lens>/NOTES.md` — 22,300 lines.

1. The answer to your question

Yes, people day-trade MNQ profitably. No, nobody with a verified record makes \$1,000 a day on \$30,000, and the reason is arithmetic rather than skill. The best documented intraday futures records pay **\$15-\$125 a day on a \$30,000 account**. The single closest twin to you I could find anywhere — Pau Perdices, a publicly-tracked, commission-net, discretionary intraday micro-gold + micro-Nasdaq account — made **\$7,163 net on \$10,000 in 8.3 months** through a **-21.2% drawdown**: about **\$124/day scaled to \$30k**. The top 500 out of ~450,000 Taiwanese day traders, measured out of sample the following year and net of all fees, earned **+37.9 basis points a day = \$114/day on \$30k**. That is the 99.9th percentile of the largest audited day-trading dataset in existence.

\$1,000/day net on MNQ is 500 index points a day, every day. MNQ's average whole-session range is 387 points and its average first-90-minutes range is 267 points — so one lot is arithmetically impossible, it is 129% of the entire daily range. The honest grid is required gross edge per round trip = $500 / (\text{lots} \times \text{round trips}) + 1.25$ points. At 10 lots and 6 round trips a day that needs **9.58 points of gross edge per round trip**. The measured gross edge available on MNQ is **1.05-1.50 points** — our own five-method result, independently confirmed from outside on the exact contract by Mesfin (2026), who tested 14 signal families on 947 MNQ sessions and found a ceiling of 0.07-1.50 points. **Every cell of that grid exceeds the ceiling, and the friction term alone is 1.25, so no amount of size or frequency can push the requirement below it.**

And here is the reconciliation with the forums, which I think is the single most useful thing in this report: the posts are probably true about the dollars and false about the capital. Topstep's own published 2025 statistics say **51.8%** of persistent participants pass an evaluation at least once — so the posters are real — but only **0.71%** ever reach a live funded account. A man "making \$1,000 a day" is nearly always doing it on a ~\$150,000 prop account, which is **0.67% a day**. You are asking for **3.33% a day**. Same edge, five times the capital, and the capital is not his.

What is genuinely on the table. Our own gross edge, at a sane size, independently produces **\$15-\$30/day** — which lands within a dollar of the broker-verified intraday index records. Push to 10 lots with a 4R average winner and the arithmetic permits **\$400-500/day**, at the price of a ten-loss streak costing 17-27% of the account. **So \$1,000/day on \$30k is not impossible in arithmetic — it is impossible in drawdown.** Capital required to run it survivably: **\$300k-\$900k, centred around \$400k**.

\$15-\$125PER DAY ON \$30K —
VERIFIED INTRADAY
RECORDS**500 pts**MNQ POINTS/DAY THAT
\$1,000 REQUIRES**1.05-1.50**POINTS OF GROSS
EDGE ACTUALLY
AVAILABLE**0.71%**OF PROP ENTRANTS
REACH LIVE FIRM
CAPITAL**The verified records, in dollars per day on a \$30,000 account**

WHO	GRADE	WHAT IT IS	ANNUAL	\$/DAY ON \$30K	RISK TAKEN
P&LFidelity (FundSeeder, broker-verified)	A–	Index futures, no overnight	12.3%	\$15	5.7% vol · FundSeeder score 87.1
CapitalCross (FundSeeder, seeded)	A–	Index futures mean reversion	22.5%	\$27	6.5% vol
K. Michael Wallace (World Cup Advisor)	B+	ES/NQ/MES/MNQ, ~1-day holds	22.6%	\$27	–1.33% worst drawdown
AsymmetricOverbought (FundSeeder)	A–	Intraday index futures, flat overnight	27.2%	\$32	23.7% vol
EquityPortfolio (FundSeeder, CH)	A–	Trend, built alongside a day job	55.5%	\$66	43% vol
Best lifetime compound, 191 CTA programs	A–	AutumnGold database	~50%	\$60	top 15 includes –93%, –88%, –66%
Barber et al. top 500 of ~450,000	A	Ranked prior year, measured next year, net of fees	+37.9 bps/day	\$114	99.9th percentile
Pau Perdices (World Cup Advisor)	B+	Discretionary intraday MGC + MNQ + MES , \$10k account, 8.31 months	+71.6%	\$124	–21.2%
Jigsaw leaderboard July winner	B	Best of 307 traders, one month, live account	—	~\$250	ramped 4→10 lots inside the month
THE TARGET	—	MNQ + MGC, first 90 minutes	~840%	\$1,000	—

Not one professional intraday index futures programme is an income business. R Best LLC (100% intraday ES + machine learning, 12 years, \$22M) is roughly flat 2022–2026. Deep Field "Intraday Crisis Alpha" lost money in four of five years. QTS Tail Reaper — Ernest Chan's own firm, intraday ES — did –23.0% then –29.6% with a –67.6% max drawdown and AUM down to \$233k. Comhla "Artemis 3X", a registered CTA's \$35k-minimum ES/MES/NQ/MNQ programme, has a **negative lifetime compound return at a –66.4% drawdown.**

The grid that settles it

Required **gross** edge in MNQ points per round trip to net \$1,000/day, at 1.25 points all-in. Our measured ceiling is 1.05–1.50 points; cells above 1.50 are unreachable.

LOTS	2 RT/DAY	4 RT/DAY	6 RT/DAY	10 RT/DAY	20 RT/DAY	FRICTION/DAY AT 10 RT
1	251.25	126.25	84.58	51.25	26.25	\$25
5	51.25	26.25	17.92	11.25	6.25	\$125
10	26.25	13.75	9.58	6.25	3.75	\$250
20	13.75	7.50	5.42	3.75	2.50	\$500
30	9.58	5.42	4.03	2.92	2.08	\$750

The lowest cell — 30 lots, 20 round trips — still needs 1.4–2.0× the measured ceiling, costs **\$1,500/day in friction (5% of the account, daily)** and runs **59× leverage**. Run it forwards at the top of the measured range (1.50 gross, so +0.25 net per round trip): 5 lots × 6 RT = **\$15/day**; 10 × 6 = **\$30/day**; 20 × 10 = \$100/day; 30 × 20 = \$300/day at 59× leverage. **Two completely independent routes — the audited records and our own tape — converge on \$15–\$32/day, and neither requires believing the other.**

Drawdown is the binding constraint, not margin and not return. MNQ day margin is ~\$464/lot, so margin permits **64 lots** on \$30k — it will not stop you. The 127 genuinely-profitable Brazilian day traders in the largest index-futures census ran daily standard deviations **6.2–8.4× their own daily mean**. At a \$1,000/day mean that is a **\$6,200–\$8,400 daily SD = 21–28% of a \$30,000 account**: a two-sigma day is –41% to –56%, a three-sigma day is the account. The prop industry's own rails scaled to \$30k are a **\$600 daily loss limit and a \$1,200 trailing drawdown** — so \$1,000/day is **1.67× the daily loss limit and 83% of the entire drawdown allowance** the industry gives an account this size.

And gold cannot substitute. \$1,000/day = 100 MGC points against an average first-90 range of 38.6. The measured book supports about **5 lots**: a 10-lot MGC order costs a median \$3.60–\$4.60/lot above the best ask (p90 \$17.80–\$21.70) — more than gold's entire spread — taking all-in to \$12–13/lot against the \$4.50 constant. Ten MNQ lots cost \$0.35/lot. **Not executable in gold at any size.**

2. Where the schools agree

This is the heart of the report. Unaudited practice cannot offer proof, but it can offer **convergence**: when order-flow traders, price-action traders, market-profile traders, prop-firm risk departments, academics and a 42-year trading championship independently say the same thing, that is the strongest evidence this kind of material can give. I graded convergence by independence — three DOM vendors who host each other's videos are one tradition, not three samples; likewise Zarattini, who is one research group behind most of the quantified tier, and the price-action schools, who share Brooks' vocabulary.

#	THE AGREEMENT	INDEPENDENT TRADITIONS	QUANTIFIED?
1	Nobody forecasts direction. Every profitable method routes around it: Crabel rests both stops and lets the market pick; Zarattini takes the sign of the first 5-minute body; Fisher demands dwell rather than prediction; calevonlear: "Direction is meaningless, afterwards is what matters." And not one verified track record in the whole sweep describes predicting direction — every single one names its edge as flow, positioning, an event, or a tail.	9 of 9	Yes — NQ first-break direction is 49.6%/50.2% on n=1,670 sessions
2	Size is an OUTPUT of a demonstrated live record, never an input. Brooks' ladder: \$10/day → 3 ES contracts over a year. The Jigsaw leaderboard winner: 4→10 lots inside one month. Axia: 3→300 lots over three years. The prop ladders: ~1 mini per \$10,000 of retained profit, with 10 active trading days per tier and no skipping , while demotion is automatic and same-day. Our "live at ~\$30k" plan implies fixed size on day one. Nobody in this literature does that.	7, unanimous	Yes — published ladders
3	Stops and targets go at STRUCTURE, not at a distance. Valtos opens an article describing this desk exactly: "most traders use arbitrary stop levels. Maybe they have back tested their system and see that the average loss is 10 ticks, so all of a sudden their stop is 10 ticks on every trade... Wouldn't it make more sense to have a stop level based on what the market is telling you?" Brooks: "the first target is the previous swing low." Beggs: "any trailing stop must be at a place where you consider the trade idea is no longer valid. ANY CLOSER IS TOO CLOSE."	7	Partly
4	TRADE SELECTION carries the edge; the trigger does not. The only finding in the entire sweep with a published dose-response. Same entry, same stop, same exit, only the universe changed: 5-minute ORB on 7,000 US stocks = Sharpe 0.48 (loses to buy-and-hold); filtered to opening-range relative volume ≥100% and the top 20 names = Sharpe 2.81 . Per trade, net of commissions: RelVol <100% = -0.02R (negative), >100% = +0.08R, >30x = +0.38R.	5	Yes, with a dose-response
5	The payoff is the defect, not the hit rate. Four unrelated profitable books sit at 18–25% win rate, same as ours. The difference is entirely the average winner.	5	Yes, twice
6	The first 1-2 hours are where the money is — and our own tape agrees: the first 90 minutes after the cash open contains 72.2% of MNQ's entire RTH range on ~42% of the volume, so per minute of attention it is 4.3x as productive as	6	Yes — measured on our own data

the remaining five hours. Jigsaw's June leaderboard winner: "the bulk of the profits were made from just 2 hours of the day."

7	KILL THE FIXED TARGET. The strongest quantified convergence in the sweep: all four sources have samples. Zarattini's 2-D stop×target grid found the optimum at the tightest stop tested with no profit target at all , held to the close. Wu et al. on Taiwan index futures: "the introduction of a profit target is detrimental for overall profitability." Carver refuses targets on principle. And our own tape already agrees — TRAIL earned +\$186 and TARGET_200 +\$197 against +\$62.69 for the tight target.	4	All four quantified
8	The stated cause of failure is taking profit too early , not entering badly.	5	No
9	A hard, un-editable daily loss limit. Every professional hard rule in the industry is on loss, never on win rate or target. QTS's published objective: "daily stop loss ≤ 5% of nominal." Collective2's C2Star certifies purely on drawdown while the return bar is merely "beat the S&P by +1% over rolling 60 days". Topstep doubles your payout cap if you voluntarily add a daily loss limit. Their own stat: "Over 63% of traders have lost account in a single day."	6	Partly
10	★ The "second attempt" is the highest-probability entry — Brooks' H2/L2, Volman's Second Break ("arguably no higher probability of a winning trade in the market"), PATs' second entry, Beggs' complex pullback, four schools converging on one pattern. This is the largest convergence in the sweep with the WEAKEST evidence: five or six schools, zero samples, anywhere. Shadow-book candidate only. Never a change to anything live.	5-6	No samples at all

The three cross-lens resolutions no single source could see

(a) Your 90-minute habit is vindicated by the instrument — and the same measurement reverses the recommendation

Al Brooks asserts the day's high or low occurs within the first 90 minutes about **90%** of the time. That claim is the load-bearing justification for trading only the open, and its one public check (TradingView, n=29 ES days) got **59%**, not 90%. We measured it on our own MNQ tape and **Brooks is right: 90.0%** — the public refutation failed on a definition, measuring whether the range survived rather than whether an extreme occurred. But the same measurement then reverses the advice:

MEASURED ON OUR OWN MNQ TAPE (239 RTH SESSIONS, DST-CORRECT)	MNQ	MGC
Share of whole-session range contained in the first 90 minutes	72.2%	67.7%
Share of RTH volume in that window	~42%	—
Days where the window contains ONE of the day's two extremes	90.0%	—
Days where it contains BOTH	10.4%	20.8%
Days that set an extreme in the 15:00–16:00 ET hour you never see	37.2%	22.7%
Days where the first extreme arrives inside 30 minutes	53.2%	—

So the window is structurally **continuation geometry** — price has just left one extreme and is travelling toward another that prints in the afternoon — and **stopping at 11:00 ET abandons an unfinished range nine days in ten**. Worse: **your window opens on the single most expensive five minutes of the MNQ day**. 09:30–09:35 ET has a median 3-tick spread, \$1.36 a crossing against \$0.86 at 10:15, and 3 contracts a side instead of 5 — widest bars, widest spread and thinnest book are all the same five minutes. Meanwhile MNQ's spread is tightest 16:00–19:00Z, which is the part of the day nobody looks at.

(b) Four literatures 34 years apart are measuring one latent variable: "is today busy?"

Zarattini's opening-range Relative Volume (2024, dose-response $-0.02R$ / $+0.08R$ / $+0.38R$) is Toby Crabel's "Early Entry 2" from **1990** — first-bar volume against the same bar over the previous 15–20 days. It is also Dalton's Open Drive (17% of sessions, 62.5% continuation, the only opening type in Market Profile that survives a 2:1 test) and the ORB width tiers (wide opening ranges $>0.6 \times \text{ATR}$ continue 74.2% on NQ against 63.6% for narrow ones). **Four traditions, one measurement**. Use RelVol: it is the only version that is objective and has a published dose-response. Corollary worth saying out loud: the Market Profile open-type taxonomy is probably not a separate research programme at all, just the same measurement with a subjective classifier bolted on.

(c) The MGC contradiction resolves into a decision

The order-flow lens concluded gold is the friendly instrument — a 30-tick MGC stop carries only 15% friction, and it is the only configuration where \$1,000/day looked arithmetically reachable. The instrument lens, having actually walked the book, found MGC's practical ceiling is ~5 lots at \$12–13/lot all-in. The instrument lens wins because it measured rather than assumed the constant scales. **The synthesis nobody states: MGC is the friendlier instrument PER LOT and the worse instrument AT SIZE**. So gold is where to test tight-stop, level-anchored methods at 1–3 lots, and MNQ is the only instrument that can ever carry income size.

★ A consequence that reprices a lot of existing work: the paper engine's fabricated 0.1% adverse fill is **\$29.39/lot on MNQ against a measured real 10-lot book-walk of \$0.35/lot — 84× punitive** — but on MGC it is \$4.37 against \$4.60 measured, i.e. roughly realistic. **MGC's paper P&L is far more trustworthy than MNQ's, and any correction must be per-instrument**. A single global fudge factor would be exactly the "right number beside a wrong one" bug.

Where they disagree, with my verdict

THE QUESTION	THE TWO CAMPS	VERDICT
Move the stop to breakeven?	Mack does it on every scalp; Volman says it is "simply asking for an early exit"; Carver's theory says never	NO. Every measured source plus theory is against it. Every voice for it comes from a school whose own founder says "back testing is bunk."
Trailing or fixed stop?	Sinclair's simulation has no autocorrelation and favours fixed; Zarattini's max(band, VWAP) trail doubled a measured Sharpe 0.61→1.24 on 17 years of real data	Genuinely open , and it is really a question about whether trends exist at the horizon traded.
Structural or ATR stops?	Practitioners say structure; the quantified side uses $k \times \text{ATR}$	The quantified side wins on evidence — but the winning distance-stop is itself structural (the first 5-minute candle's extreme). The untested family is the hybrid: nearest structural level beyond $k \times \text{volatility}$.
Limit on the retest, or stop on the break?	Five unconnected forum posters say limit on the retest; Valtos measures ~50% fill and says the missed half is the strong half	It does not buy a better price — that is adverse selection, and it is exactly why our own passive entry died. It buys a much tighter stop, which is a different hypothesis and worth testing as one.
How long is the opening range?	Market Profile's constants are 30 and 60 minutes; Zarattini gets 5m Sharpe 2.81 vs 30m 0.21; Tsai finds the best probe on E-mini Nasdaq is one minute	1-5 minutes on US instruments. Two independent programmes agree, and Market Profile's two central time constants are the two worst cells on the sample.
Tighten or widen the stop?	Everyone quantified says tighten	Tighten — but this lever is already spent here. Our \$49.59 stop is 24.8 MNQ points $\approx$ 12.4% of a 200-point ATR14 , already inside the 5–10% zone Zarattini's grid preferred, and MNQ friction floors any stop at ~20 ticks. Stop tuning stop width.

3. The method map

One section per part of the trade, with mechanics specific enough to implement, and names attached. Evidence grade in every row: **A** verifiable/peer-reviewed · **B** a real sample, unaudited · **C** assertion or marketing.

3.1 Trade selection and when NOT to trade — the category we have never built

Every gate we own is a direction or regime trigger. We have never built a selection filter, and this is the source of essentially every profitable result in the sweep.

RULE	MECHANICS, EXACTLY	THE NUMBER	GRADE
Opening-range Relative Volume Zarattini / Barbon / Aziz 2024; Crabel 1990	RelVol = volume in the first 5 minutes today ÷ mean first-5-minute volume over the prior 14 sessions, same contract. Require $\geq 100\%$. In stocks, also take only the top 20 names.	Sharpe 0.48 → 2.81 −0.02R / +0.08R / +0.38R by bucket	B
NR4 / NR7 / Triangle Zarattini et al. on 2,620 SPY days; patterns from Crabel 1990	All measured on DAILY bars before the open, conditioning the whole session. NR4 = today's daily range is the narrowest of the last 4. Triangle = high below the prior 2 highs and low above the prior 2 lows.	Unconditional 12bps, Sharpe 1.7 NR4 22bps, t=5.14, Sharpe 3.2 NR7 t=3.07 · Triangle t=3.19	B
Do NOT trade momentum after a trend day	Mechanical trend day = open below the 15th percentile of the day's range, close above the 85th, range above the 14-day average.	−2 bps, t=−0.24, Sharpe −0.3 n=215 of 2,620	B
Nulls, so nobody re-derives them	Inside Day and Outside Day as conditioning variables.	t=0.85 and t=1.05 — nothing	B
Volatility regime FOUR independent programmes	Condition entries on VIX/ VIXN or realised-vol state measured before the session; stand	Sharpe ~1.5 above VIX 6, ~3.5 above VIX 40 Li/Sakkas/Urquhart: US R ² 0.84% in expansions vs 7.53% in crisis Holmberg: "ORB is basically LONG VOLATILITY"	A

aside in the middle.

Two horizons must AGREE Baltussen et al., JFE 2021	Go long only when the first-half-hour return AND the rest-of-day return are both positive; short only when both are negative; do not trade at all when the signs differ.	Success 0.55 → 0.61 Sharpe 1.07 → 1.60 while raw return FALLS 6.86%→5.47%	A 60+ futures, 1974–2020, tick data
5-day RSI as a dealer-gamma proxy	Prior close's 5-day RSI of the index. Low RSI ⇒ dealers short gamma ⇒ hedging amplifies intraday trends.	$\beta = -3.25$ bps per RSI point, p = 0.001 RSI 30→70 costs ~130bps/day	B
Beggs' Rule of Three — the only halt rule in the sweep that is neither P&L-based nor direction-based	RED conditions (bad feed, no charts, under 6 hours' sleep) ⇒ stop. Three concurrent AMBERs ⇒ stop.	—	C
The event buffer	No new entries within ~2 minutes either side of a high-impact release — the convergent rule across every prop firm's funded-account documentation. For FOMC, bench 17:55–18:40Z to cover both the 14:00 statement and	Fed survey: announcement-window R^2 only 0.31 at best Lucca & Moench: announcement-to-close return "essentially zero"	A

the 14:30 press
conference.

**Dalton's
default**

"No trade is a
legitimate
trade. When
confidence is
low you
generally want
to stay
uncommitted
so that you can
execute if the
confidence
level changes."



Banked nulls in this category so no night is wasted on them: Firich (2012) measured $P(\text{trend day tomorrow})$ at 9.50% / 9.94% / 10.39% depending on today's type, against an unconditional **9.54%** — day type carries zero next-day information. And the Initial Balance is broken on **73.7-78.3%** of days, so **any gate that fires on an IB break is a clock, not a signal** — the identical shape to our own withdrawn tunnel_watch claim, arrived at independently in 2012. Trend-day frequency triangulates four ways at **8-10% of sessions** (Dalton "<10%", Firich 9.54%, Zarattini 8.2%, vovoperador "about 10%").

3.2 Entry

METHOD	WHO	MECHANICS	GRADE
5-min ORB	Zarattini & Aziz	If the first 5-minute candle is UP, buy-stop at its high; if DOWN, sell-stop at its low; doji = no trade. Stop at the candle's opposite extreme (= exactly 1R). No target. Hold to the close.	B
Kaufman volatility breakout — the cleanest replication target in the sweep	Kaufman 2013, implemented by Zarattini & Pagani	Bands = session open $\pm 0.5 \times \text{ATR}(14)$ of daily bars, fixed all session. Enter on a bar CLOSING outside. The stop is a return to the session opening price — one level, known at the open, never moves. No target. Flat at EOD. Decisions only at HH: 00/15/30/45. 19 years SPY: CAGR >13%, Sharpe 0.87 net of IBKR fees , with no regime filter at all.	B
The "Noise Area" — a when-not-to-trade model before it is an entry model	Zarattini / Aziz / Barbon	For each time-of-day HH:MM, σ = mean over the prior 14 days of $ \text{close}(\text{HH:MM})/\text{open}(09:30) - 1 $. Bands = $\max(\text{today's open, yesterday's close}) \times (1 \pm \sigma)$. Inside the band, hold nothing. Close above the upper band = long. Decisions only at HH:00 and HH:30 — a deliberate anti-overtrading clock. Flat at the close. It never forecasts direction; it takes the sign of an already-realised displacement.	B
The second attempt	Brooks (H2/L2), Volman (Second Break), Mack, Beggs	Do not take the first break of a level; take the second. Volman: "arguably no higher probability of a winning trade in the market."	C — no samples
The retest, on a resting limit	Fisher's ACD; five unconnected forum posters	Break the range, then enter the retest with a limit rather than the break with a stop. "Using buy and sell stops is usually entering at worst possible price." Stop beyond the breakout bar; target structure at 1.5–2R.	C
Fast alpha as an EXECUTION overlay — the one idea aimed squarely at our shelved OFI work	Zarattini & Pagani 2026; Heston/Korajczyk/Sadka, J. Finance 2010	Do not trade a short-horizon signal. When a slower strategy fires a long, delay the fill until one adverse 5-minute bar has printed , then buy; apply the same delay to the stop. Add an urgency timeout to force the fill. The standalone signal is worthless after commissions; as an overlay it lifts CAGR ~200bps and Sharpe 0.87→0.99, net , with zero added round trips .	B
ICT Silver Bullet	ICT (second-hand; his own hosts are bot-walled)	10:00–11:00 ET window — inside your existing watch. On MNQ that translates to a 5–8 point stop and a 10–15 point target, i.e. 8–12% friction. Cost-viable.	C

3.3 Stops — three objects we do not have

OBJECT	MECHANICS	WHY IT MATTERS HERE
The structural stop Valtos, Orderflows	The stop sits where the premise is falsified — the far side of the initial balance, the re-entered value-area edge, the origin of the day's largest stop run — not at an ATR multiple.	★ A distance cannot encode an invalidation. That is precisely why a 430-configuration sweep over stop distances found a shallow optimum and reported "all configurations $\leq$ holding."
The two-tier stop Valtos	A soft 2-3 tick stop that is not in the market, whose trigger is to LOOK at flow: "I am interested in watching how the market REACTS when it gets to my mental stop as opposed to actually getting out." Under it, a hard worst-case stop that is in the market. He publishes the paired ZB trades: hard 2-tick stop → stopped, then it reversed to target ("the operation was a success but the patient died"); mental stop → 3 ticks against, then target.	We have only Tier 2. Every stop-out is a hard stop with no machinery between "in" and "stopped" . And we are architecturally ready: <code>gazbot7-rider-peak-watch</code> is already the right shape — a 1 Hz read-only loop that raises a question with the hard stop untouched underneath.
The MAE-conditional, pace-scaled time stop Valtos	"If the market doesn't go to my take profit or stop level after a certain amount of time, the reason for getting into the trade has most likely diminished." Scaled to the instrument's pace, and conditional on whether the trade has ever gone against you. He publishes a case where it cost him 19 of 20 ticks and defends it distributionally.	We have <code>DAYTRADE_FLAT_CLOCK=20min</code> — so the mechanism exists but is blanket rather than per-signal and MAE-conditional. MNQ is one of the fastest liquid futures, so its clock should be short.
The scratch, as a third outcome Valtos	"If you are trading with indicators it almost feels like an ALL OR NOTHING way of trading. You will either get to your profit target or get stopped out. THAT ISN'T THE WAY TO TRADE."	Our 536 resolved trades all ended at one of two pre-set prices. 400 stops is 400 instances of learning last.

The theorem that reclassifies our biggest null. Kaminski & Lo (J. Financial Markets 2014) prove that **under a random walk a stop-loss ALWAYS reduces expected return**, and adds value only in the presence of momentum. We have established by five independent methods that MNQ direction is ~ 0.50 at every horizon. So **400 stops at – \$49.59 against 136 targets at +\$62.69 is not a bad stop level — it is the theorem being paid**, and our "430 exit configurations all $\leq$ holding" result is an unwitting empirical confirmation of a published theorem. That moves it from a disappointing null to a **settled question**: further unconditional exit-lab work can stop. The one question the theorem leaves open is the conditional one — do stops add value only in the subset where momentum plausibly exists?

And the cost arithmetic kills the whole tick-scale DOM practice on MNQ. At \$2.50 all-in a 1-tick MNQ stop is 500% friction and a 5-tick stop is 100%. The DOM school was developed on ZB/ES/Bund where a tick is \$12.50–\$31.25 on a very deep book. **Minimum coherent MNQ stop is ~20 ticks (5.00 points) for $\leq 25\%$ friction, ~50 ticks for $\leq 10\%$.** Mack's 4-tick ES scalp translated literally to MNQ is 1 point = \$2.00 against \$2.50

of cost — structurally negative before any win rate at all. On MGC at \$4.50 all-in, **no scalping school in the sweep survives translation**; only the wide-target structural shapes do, which matches our own finding that gold's runs are grinds and the exit was the cap.

3.4 Targets and management — where our actual defect is

BOOK	WIN RATE	AVG WIN	AVG LOSS	PAYOFF	EXPECTANCY
Zarattini & Stamatoudis, blind anonymised charts, 1,580 trades	18%	+10.10R	−1.02R	9.9 : 1	+1.03R
Zarattini & Aziz 5-min ORB on QQQ, 1,795 trades	24%	~3.7R	−1.00R	~3.7 : 1	+0.13R
"Fenix", Jigsaw public leaderboard monthly winner	~50%	\$619	\$489.05	1.266 : 1	+\$64.98
GAZBOT V7 live book, 536 resolved	25.4%	\$62.69	\$49.59	1.264 : 1	−\$21.07 (−0.425R)

★ **Our win rate is completely normal for a profitable intraday method — 18%, 24%, 24%, 25.4%.** Our payoff is not. And note the third row: Fenix's payoff ratio is **1.266 against our 1.264 — identical to three decimal places** — and he cleared the 44.2% break-even hit rate by six points while we miss it by nineteen. **That means "we need 3:1" is only one of two exits, and the empirical winner took the other one.** Put most usefully: we do not need more winners, we have 136. If we had taken only ~272 trades and those same 136 had won, we would be Fenix. **The problem is the ~264 trades we should never have entered — which is a trade-selection statement, not an exit statement.**

The corroborating frontier: FTMO publishes per-account post-mortems of traders who actually got paid, with win rate and average reward:risk read off its own broker data. Twenty complete pairs, and the shape is **bimodal with a hole in the middle**: every paid trader at win rate $\leq 46\%$ carries R of 1.58–5.59, and every one at $\geq 55\%$ carries 0.65–1.83. Minimum edge index $w \cdot R / (1 - w)$ observed across all twenty is **1.295**. **Ours is 0.430. Not one paid trader sits where we sit.**

WHAT A STOP IS ACTUALLY FOR — THREE INDEPENDENT DEMONSTRATIONS	HIT RATE	PAYOFF	SHARPE	SKEW
Sinclair, 10 years SPY: buy-open/sell-close, no stop	54.3%	0.85	0.066	—
...same trades with a vol-scaled stop (half the 20-day SD, reset each morning)	47.0%	1.18	0.298	—
Zarattini SPY 2007– 2024: trailing stop at the opposite band	54%	—	0.61	–1.24
...same entries, trail at max(current band, VWAP)	43%	—	1.24	+1.29
Han/Zhou/Zhu, 88 years of momentum, 10% stop	—	—	more than doubled	worst month –49.8% → –11.4%

★ A stop SELLS hit rate to BUY payoff. We have paid the hit-rate price and never collected the payoff, because a \$62.69 target truncates the exact right tail the stop bought. The operational consequence is a rule for the badcall ledger: **a falling hit rate is not evidence of a worse exit**. Judge exits on payoff, skewness and worst-day — never on hit rate.

3.5 Sizing

METHOD	MECHANICS		THE NUMBER	DOES IT WORK AT \$30K?
Volatility targeting Zarattini; theory from Harvey et al., JPM 2018	shares = AUM × min(4, $\sigma_{\text{target}} / \sigma_{14\text{-day}}$) / open, with $\sigma_{\text{target}} = 2\%$ daily. Size fixed at the session open, unchanged intraday.	The identical 7,668 trades at the identical \$0.09/share went 380% → 1,985%		No. At \$30k the multiplier rounds to {0,1,2} lots. The single largest documented improvement in intraday index trading is unavailable at this account size. The reachable half is binary participation — 1 lot on normal-vol days, 0 on high-vol days — which gets the tail reduction and none of the Sharpe.
Breadth Carver, measured over 37 futures	More instruments, same signal.	1 instrument SR 0.35 · 3 → 0.50 · 5 → 0.57 · 8 → 0.61 · 37 → 0.70 discreteness penalty ~20% of Sharpe at max 1 lot, ~5% at 2, zero at 3-4		Yes — and it is the cheapest Sharpe on the menu. 1→8 instruments is +74% Sharpe with no new signal. His explicit \$30,000 portfolio is FIVE instruments — written in 2016, before the micro complex dissolved the "can't hold less than one contract"

problem he was solving. Breadth here means MES/MYM/M2K/MCL/M6E, not more gold.

Size as a function of retained profit Topstep, Apex ladders	Daily loss limit $\approx$ 10% of cumulative profit; $\sim$1 mini per \$10,000 of cumulative profit; promotion needs 10 active trading days per tier with no skipping; demotion automatic and same-day, with the tier frozen at the prior close so it cannot change intraday.	At \$30k with nothing retained: ≤ 2 minis / 20 micros = exactly Kevin Davey's "one contract per \$15,000"	Yes. Note that an "active day" is any day with ≥ 1 trade, so the clock rewards showing up — converging with the strongest finding in the base-rate literature.
Scale IN Jigsaw Rule 20; Hougaard	Three named schedules: Standard Pyramid 50/30/20 (safest, keeps average price furthest away), Equal Block, Inverted Pyramid ("quite lethal"). Control variable is average price / break-even measured in ATRs. The reframe: "as you add, the price is mathematically worse, but you have superior information — you are paying		— Untested here — we only scale out. $\triangle$ Contaminated on the paper engine (see 84x artefact).

a premium for information." And the honest admission: "your last scale-in will probably be a loser."

Risk per trade, as every school actually sets it

0.25% to 1% of equity.

\$75-\$300 at \$30k

★ **This is the whole distance between the forum post and the paper.** \$1,000/day at those rails implies **3-13R every single day** against measured per-trade expectancies of +0.13R to +1.03R at ~1.8 trades/day. To chase it you must risk ~4.3%/trade, where at an 18-25% win rate a five-loss streak is **37% likely** and a ten-loss streak (13.7% likely) costs **36% of the account**. The practitioners whose numbers survive scrutiny risk 0.25%. **Factor of 17.**

★ **One bias a bot can simply not have.** Ben-David, Birru & Prokopenya (NBER w22146) have full records from a retail FX broker and document a discontinuity in trade size at exactly zero past-week return worth 12-17% of a standard deviation (~55% for trade count) — **with no corresponding discontinuity in future performance**. Same effect in CBOT floor traders and Taiwanese market makers. If size is a pure function of (equity at session open, current instrument volatility) and is **frozen for the session**, the bias cannot exist. That is free.

3.6 Daily risk limits — what the industry actually enforces, scaled to \$30,000

LAYER	RULE	AT \$30K	DETAIL THAT MATTERS
1 — account floor	Trailing max drawdown, 4% at every firm and every size	\$1,200	Trails the closing balance, never trails down, locks permanently once it reaches the starting balance , and is evaluated on unrealised P&L.
2 — day floor	Daily loss limit, ~2% (Apex uses exactly half the drawdown at all four sizes)	\$600	On total equity including open positions. Hitting it is not a failure , it is a forced stand-down until the next session — and it must be un-editable for the session .
3 — the upside stop nobody advertises	A Topstep performance coach's own rule: quit when up more than \$300 on a \$50k Combine = 0.6% of account	\$180	Plus a day-level give-back ratchet: "once up \$1,000 I will not go home with less than \$800. PERIOD", the floor moving up with each new high. The sharp counter-argument, worth testing: "Don't limit profits unless you consistently give them back and if you do then you have a TIME OF DAY problem, not a max profit problem."
4 — consistency	Best single day ÷ net profit ≤ 20-55%	—	★ A big day is priced as a NEGATIVE signal by every firm that observes hundreds of thousands of futures accounts.

★ **The non-obvious interaction, and it is important: a low-hit-rate design needs a WIDER daily loss limit than the industry default.** At our 25.4% hit rate, four losers in a row happen 31.6% of the time, ten in a row 5.6%, and sixteen in a row is the 99% point ($0.75^{16} = 0.01$). Acrary's arithmetic puts the **trading cycle to a new equity high at 48 trades**. Topstep's 2% limit is calibrated on high-hit-rate scalpers and would **halt a convergent low-hit-rate design out of its own edge**. Calibrate the two rails together — 3-4R, not 1R.

And a free prerequisite before building any of it: is our worst day a large multiple of our median losing day? If losses are single-session-dominated, a hard daily loss limit is the highest-value change available and needs no edge at all. If it is a slow bleed, the limit buys nothing — and that itself is the finding: our failure mode is not the industry's. We have never looked at the shape of our own daily loss distribution.

4. The ten things this desk has never tried

The arithmetic that ranks them

136 winners at +\$62.69 = **+\$8,526**. 400 losers at -\$49.59 = **-\$19,836**. Every worthwhile finding in the sweep reduces to one question: **how do you get the loss pool under \$8,526 without touching the winners?**

ROUTE	LEVER	MOVE NEEDED TO BREAK EVEN ALONE
A. Cut loss SIZE	trade management, stop architecture	avg loss \$49.59 → \$21.31 (-57.0%)
B. Cut loser COUNT	trade and day selection, stand-aside rules	400 → ≤171 losers (-57.3%)
C. Raise winner SIZE	remove the target, let the tail run	avg win \$62.69 → \$145.85 (+133%)

★ **The two 57% figures are the same number reached from opposite ends, and they COMBINE MULTIPLICATIVELY.** 30% fewer losers × 40% smaller losses = $0.70 \times 0.60 = 0.42$ of the loss pool = **\$8,331, which is under \$8,526** — a profitable book with zero change to entry direction, to the hit rate of the trades we keep, or to the winners. **We do not need a better entry. We need 30% fewer entries and 40% smaller losses — two independent projects, neither of which requires a direction forecast.** Route C we have half-measured already (TRAIL +\$186, TARGET_200 +\$197). Routes A and B we have never systematically attempted at all.

Four framing rules — each of which invalidates a class of work already done

RULE	WHY
R1. Only LARGE-GAP comparisons are resolvable on 879 trades.	MacLean, Thorp & Ziemba: telling a 1.0% edge from a 1.1% edge takes two million trials . ★ This retroactively explains the 430-exit-configuration null — the study was unresolvable by construction, not the world being flat. Every test below is a 2-, 3- or 4-cell factorial with a large expected gap. Any future sweep with more than 8 cells on the live book should be refused at design time.
R2. A falling hit rate is not a worse exit.	Three independent demonstrations (§3.4). Drop hit rate as an acceptance test; judge on payoff, skew, worst-day. Our instinct will read a hit-rate drop as a regression, so this needs to be a written line in the badcall ledger.
R3. Every multi-lot P&L number is contaminated, PER INSTRUMENT.	The paper engine's 0.1% adverse fill is \$29.39/lot on MNQ against \$0.35 measured (84× punitive) and \$4.37 against \$4.60 on MGC (realistic). Any sizing or scale-in test on the paper engine measures the simulator, wrong in opposite directions on the two instruments.
R4. The measuring instrument itself is unverified.	IBKR's own documentation: streaming market data "is not tick-by-tick but consists of aggregate snapshots taken several times per second" (depth, by contrast, is "sent without sampling nor filtering"). Jigsaw's V6.1 changelog, 2025-12-05, lists "Fixed MAE/MFE (not for IB — that's down to a data restriction)" — a commercial platform that computes MAE/MFE for eight other feeds explicitly cannot on IBKR. This does not overturn the direction nulls, but every bar, ATR, ER, VWAP and MAE we own came from aggregated snapshots with an inferred aggressor side. Cost to close: ~\$57/month (Sierra Chart Package 11 + Denali CME+COMEX with depth) ≈ 12 MNQ round trips, and it is a go-live purchase because non-pro exchange rates require a funded account. Nobody had priced it.

The ranked queue

#	WHAT	THE TEST THAT SETTLES IT	DATA / VERDICT
1	The MAE distribution, conditional on eventual outcome. Maximum Adverse Excursion is absent from our entire research history — and nearly absent from the academic literature too (zero relevant OpenAlex hits; it lives in Sweeney's 1996/97 books and vendor software). It is the one measurement that decides whether 400/136 is a stop problem or a target problem, and we have been arguing that question without it. Three lenses nominated it independently.	For all 879 live trades plus the shadow book, record each trade's worst unrealised drawdown. Plot the MAE distribution separately for eventual winners and eventual losers. Place the candidate stop just beyond the percentile that keeps 90-95% of eventual winners alive. Δ Compute the race, not "reached N R" — our own MFE trap. Δ Per R4, report the percentile as a lower bound.	Yes — 879 trades + 13 months of 1-min bars, all on disk. No new data, no P&L model, no direction forecast.
2	Remove the target. A 2x2 factorial, not another 430-cell sweep. Four quantified sources agree and our own tape already half-agrees.	Cells: (a) stop only, no target; (b) target only; (c) both — status quo; (d) neither, exit at 20:40Z. Pre-register the published prediction: (a) shows a lower hit rate and a much higher payoff than (c); Zarattini's independent 2-D grid found the optimum at the tightest stop with no target held to the close; Wu et al. found the target strictly detrimental. Judge on payoff, skew and worst-day — not hit rate (R2).	Yes. A large-gap comparison, so 879 trades can actually resolve it (R1). Our stop is already the right width; the target is the deviant.
3	Relative-volume day selection. The largest identified gap in the desk: we have never built a selection filter of any kind. Four lenses nominated it first, it is the only finding with a dose-response, and Crabel published the same filter in 1990.	RelVol(w) = first-w-minutes volume $\div$ trailing 14-session same-window mean, for $w \in \{5, 15, 30, 60\}$. Re-score the 879 live trades and the whole 35-arm shadow book into deciles and check whether the <100% bucket is where the losses live. Add NR4 / NR7 / Triangle / prior-trend-day flags off daily bars, and a VXN bucket. Δ Roll-aware: same contract month on both sides of the ratio — the MNQ roll moves 2.7%→17.3%→66.4% across one Friday-to-Sunday boundary.	Yes — pure re-scoring of data already on disk. An afternoon's work. It is a gate, not a trade, so it costs nothing to run and cuts friction by cutting trade count.
4	Loss-scale management: the flow-versus-price matrix. All 536 resolved trades ended at one of two pre-set prices. There is nothing between "in" and "stopped."	Jigsaw's 5-state matrix on (price since entry $\times$ flow since entry), with both meters reset at entry: (1) price against, flow not supporting = a wiggle, hold; (2) both against = exit before the stop; (3) price your way, flow weak = unclear; (4) both your way = hold; (5) flow your way but price not moving = the MOST dangerous state, because your side is the trapped cohort. $\star$ State 5 is	Yes — computable from the 250ms L2 + tick record, no new data. Arithmetic: the vendor's claimed 50% loss reduction takes the book to $-\\$1,392$ (closes 88% of the gap, not enough alone); 57% is break-even; 65% is $+\\$1,582$.

	exactly the cell a naive sign(OFI)==sign(position) rule gets backwards. Honest paired test: for every historical stop-out, does flow-since-entry separate the ones that would have come back from the ones that did not? Pre-register a random-early-exit control.	The most implementable practitioner finding gets us to roughly flat and must be combined with #3.
5	Three nearly-free diagnostics that reprice everything already done. (a) Ordinal position within the day — rank all 879 trades by position in session and ask at what trade number does expectancy go negative? One GROUP BY, never asked, and a free daily trade cap if it crosses early. (b) Acrary's trading cycle as a calibrated alarm threshold instead of a feeling: 16 consecutive losers is normal at 99%, cycle to a new high is 48 trades. (c) Acrary's EDGE TEST , the formal version of "a control is supposed to lose": replace our entries with random entries of matched hold length in the same year, 5,000 Monte Carlo passes, rank longs and shorts separately per year , require the 70th percentile on both sides for 3-5 years, re-check every 3 months live and stop trading it below 70 — and critically, disable stops and targets during the test so it measures entry skill alone.	Yes, all three. (c) is stricter than anything we currently apply.
6	Day-type classification — and this is NOT the tunnel study. tunnel_watch measured intraday compression and forecast nothing. These are daily-bar, pre-open classifiers conditioning the whole session, which we have never tested. NR4 / NR7 / Triangle / mechanical-trend-day / Inside / Outside flags off daily NQ and MGC bars over 10.5 years (minutes of work), then bucket the live trades and the shadow book. Plus the Rotation Factor : Sierra Chart published the source code for the classifier the whole Market Profile school refuses to write down (+1 for a higher high, +1 for a higher low, -1/-1 otherwise, summed per 30-minute period). Test it cumulatively after each period, as it is knowable in real time, measuring P(close in the top/bottom 15% of range RF after k periods) against the unconditional rate and a label-shuffled control.	Yes. Published expectations: NR4 t=5.14, NR7 t=3.07, Triangle t=3.19, Inside/Outside NULL, prior trend day Sharpe -0.3. This predicts our own "don't bench a coiled gate on quiet tape" memory from outside.
7	Sizing — the lever never pulled, and the \$30k ceiling on it. Vol-targeted sizing turned the identical 7,668 trades from 380% into 1,985%. △ At \$30k the multiplier rounds to {0,1,2}, so the testable version is binary participation : 1 lot on normal-vol days, 0 on high-vol days. ★ Pre-register a built-in bug detector : vol-scaling should raise MNQ's Sharpe and not MGC's (equity index vs commodity, per Harvey et al.) — if MGC improves as much, the implementation is wrong.	Partly. △ R3 bites hardest here. And one audit that may invalidate an existing result: if the stop-width A/B did not vary lots inversely, it varied RISK, not GEOMETRY. This is the sweep's strongest quantitative argument for a larger account.

Also never tried: size as a function of retained profit with a time lock; fixed-ratio (square-root-of-profit) scaling, which is the right shape for a small account; and scale-IN.

8 Stop architecture — the three objects from §3.3. The structural stop, the two-tier stop, and the MAE-conditional time stop.	The measurement that turns #4 from a claim into a number: what fraction of the 400 stop-outs had a detectable premise-death strictly earlier, and what does the loss pool become then? Then the hybrid nobody has tested: nearest structural level beyond $k \times \text{volatility}$. $\triangle$ A soft-stop layer is not an order path without the usual work (two exit paths in one tick both decrementing from a stale snapshot).	Yes — and rider-peak-watch is already architecturally the right shape for the soft tier.
9 Risk rails. We have no daily loss limit, no day-level give-back floor, no account trailing floor, and have never looked at the shape of our own daily loss distribution.	The free prerequisite first (see the box in §3.6): plot the daily loss distribution. Then replay the whole book under EOD-trailing versus intraday-trailing floors at \$1,200 and count deaths. ★ MyFundedFutures states the mechanism and it is the most relevant practitioner sentence found all night: aggressive trailing drawdowns mean "Win rate holds, but average reward per trade shrinks because every position is managed defensively." That produces our exact signature — hit rate intact, payoff collapsed — from the RISK ARCHITECTURE rather than the exit rule.	Yes, and free.
10 The statistics layer. We are a multiple-testing machine — 35 shadow arms, an 86-entry sim registry, 430 exit configs, a rotating greenfield programme — that ranks on raw P&L and un-deflated t-statistics.	Adopt three tools: Deflated Sharpe Ratio with N = the honest number of configurations ever tried (hundreds, not dozens); Minimum Track Record Length, which with 879 trades and negative skew yields a number we can act on of the form "we need N more trades before a Sharpe of X is distinguishable from zero"; and PBO/CSCV instead of a single hold-out. The bar is Harvey/Liu/Zhu's $t > 3.0$, not 2.0 — "most claimed research findings in financial economics are likely false."	Yes. Applying $t > 3.0$ ruthlessly to this whole sweep, the only MNQ result that clears it is Mesfin's gap-continuation short at $t=3.23$ on $n=22$. Free tonight: compute annualised return and vol for the live book and every shadow arm — FundSeeder scores a 12.3%/5.7%-vol team at 87.1 and a 470%/86%-vol trader at 62.8.
11 Breadth — the largest structural finding, and the only one in the queue needing new data.	Carver's measured ladder: 1 instrument SR 0.35 $\rightarrow$ 8 at 0.61. Add MES/MYM/M2K/MCL/M6E as shadow sleeves. $\triangle$ Resolvable counterweight: Barber et al. found real day traders who concentrate did better — concentration helps a discretionary trader exploiting specific	Needs new sleeves, so it is the slowest item — but +74% Sharpe with no new signal is the largest number on the page.

knowledge, breadth helps a mechanical system whose edge is statistical. We are the second kind. ★ **The free half applies today:** at max position 1 lot, Carver's own prescription is a **threshold filter — trade only the strongest forecasts**. That is the same prescription as Rosa's surviving threshold version and as #3. Three independent routes to "trade less, trade the strong signal."

12 Point OFI at the FILL, not the trade. We measured OFI at $r=+0.688$ contemporaneous and -0.001 at +60 minutes and filed it as dead. **That profile is the textbook definition of a fast alpha** — and three top-tier sources say short-horizon information is worth more on execution than on direction.

On every existing entry in the live record and the shadow book, delay the fill by one adverse micro-bar (or until OFI flips sign) with an urgency timeout, and **measure the change in average fill price IN TICKS — not in P&L**. Fill quality resolves in dozens of observations; P&L needs thousands.

Yes. We have 250ms L2 since 2026-08, and our own surviving ofi-veto-grind cell ($-\$172 \rightarrow +\772) is already this mechanism. The literature is saying it was not a fluke. The only idea in the sweep that does not require predicting direction at all.

★ **Why the five direction nulls are ONE confirmation of published theory rather than five failures.** Cont, Kukanov & Stoikov show price change is linear in order-flow imbalance with a slope inversely proportional to market depth — OFI is a contemporaneous **identity**, never a forecast. Takahashi, on S&P E-mini futures at one-second resolution, finds "shocks dissipate almost entirely within a second." **We measured OFI at +60 minutes, a horizon roughly 3,600× longer than the phenomenon's lifetime.** Nobody should re-run those studies.

And the reframe that follows. $\Delta \text{price} \approx \text{OFI} / \text{depth}$ means **depth is the denominator** — the price-impact coefficient $\lambda \approx 1/\text{depth}$ tells you how far a given flow will move price right now. Jigsaw says the same thing in prose: "thicker = less volatility, thinner = more volatility... it's the changes in liquidity that causes the market to shift gears." A C-grade practitioner claim and an A-grade replicated identity converging on a mechanism is the strongest evidence in the sweep. **λ is the correct input to stop width, position size, expected exit slippage and target feasibility — all of which we currently take from TRAILING ATR. λ is the LEADING version of the same quantity, and it never asks the book which way price will go.** Takahashi corroborates operationally: around news "price impact rises, return volatility spikes, and FLOW volatility FALLS" — only possible if depth collapsed. **A fixed-ATR stop is at its most wrong precisely at the news minute, inside your 90-minute window.**

Three external claims that contradict us — re-derive them on our sample, do not dismiss them

CLAIM	THEIR SAMPLE	OURS	THE TEST
edgeful: when NQ's first-15-minute colour and its noon position versus the 09:30 open agree, the close follows ~ 87% of the time	66–76 sessions	2,600 NQ sessions — 35×	The likely trap is that by noon the move has already happened. The test that settles it is net MNQ points from a 12:00 entry to a 16:00 exit against the unconditional control, after 1.25 points of friction.
Mesfin killed 14 MNQ signal families — but charged 2.00 points round trip where we pay 1.25	947 MNQ days, 2021–25	10.5 years of NQ — ~2.5× the span	★ The 0.75-point gap is 60% of our friction and sits right in the middle of his 0.07–1.50 point ceiling. A family at 1.4 points gross is DEAD at 2.00 and ALIVE at 1.25. Re-run all fourteen at our true friction before calling any of them dead.
Mesfin's one glimmer: gap continuation short, t=3.23, +14.52 points mean net	n=22 over three years	~2.5× the span should roughly triple the sample	It failed only the $n \geq 30$ gate. It is the only MNQ result in the entire sweep clearing the $t > 3.0$ bar. PARKED, NOT REFUTED

5. What the evidence grades actually look like

Be clear-eyed about this: most of what practitioners teach is ungraded assertion, and the people who teach it often say so themselves. Mack, who sells a price-action course, states it outright: "back testing is bunk... you cannot back test our price action trading strategies." Volman published six months of day-by-day skip decisions and never states the criterion — "'Skip this trade.' Maybe so, but why?" **The skip rule is the edge and it is the part that is not written down.**

GRADE	WHAT IT MEANS	ROUGHLY HOW MUCH OF THE SWEEP	WHO IS IN IT
A	Peer-reviewed, or a full-population census, or a regulator's primary document, or a firm's own legally-exposed disclosure	~15%	Barber/Lee/Liu/Odean (entire Taiwan tape, 2.34M trader-years) · Chague et al. (full Brazilian index-futures census) · Baltussen et al., JFE 2021 (60+ futures, tick data, 46 years) · Kaminski & Lo, JFM 2014 · Boyarchenko/Larsen/Whelan, RFS 2023 · Coates & Page, PLoS ONE 2009 · Cont/Kukanov/Stoikov · the CFTC's \$920.2m JPMorgan spoofing order · Topstep's and Tradeify's own performance disclosures
B	A real, stated sample with a stated method — but unaudited, and often by an interested party	~25%	The entire Zarattini/Concretum tier (survivorship-free samples, commissions charged, methods fully specified — and two of three authors sell day-trading education) · Mesfin on MNQ · Seeck on NQ · Jigsaw's public leaderboard (a real denominator, platform-recorded, published against commercial interest) · FundSeeder's broker-verified records · the Payout Junction on-chain archive · edgeful's base rates · Kevin Davey (his self-reported 107% cross-checks to the official championship table's 106.70%)
C	Assertion, marketing, or a screenshot	~60%	Every price-action school · every DOM vendor's setup library · ICT · essentially all of Market Profile · every single "\$1,000 a day" claim I found anywhere

Three honest observations about the grading.

- **A well-labelled C is still valuable**, because it tells us what to test. The convergences in §2 are built almost entirely out of C-grade material, and they are the most useful part of the report.

- **Grade A does not exist in the forum population, and the community itself concluded that.** Across ~25 years of the two largest retail futures forums, the only named audit service (AudiTrack) was dismissed as too expensive, and traders substituted "trade live in front of people". The highest grade reachable there is B — forward-declared, dated, continuous public posting — and I found exactly **three** B-grade things in ~155 pages: GFIs1's seven years of pre-declared one-trade-a-day DAX signals; one third-party 26-session manual replay of a Camarilla MNQ method (+\$4,465 on 5 lots ≈ **\$172/day**); and edgeful's computed base rates.
- **★ Market Profile has never been peer-reviewed in forty years.** OpenAlex returns 25 works for "Market Profile Steidlmayer value area trading" and not one tests it; even the CMT Association carries it only as webinars. Its entire evidential base is one unattributed second-hand 2012 frequency study, vendor blogs, and books that sell courses.

Two hazards in the 2026 web that anyone repeating this work must know about.

(a) The freshly-minted affiliate layer. "NQ opening range breakout rules" returns a wall of near-identical new domains with plausible rules tables and claimed backtests, each fronting a backtester or prop-firm affiliate. Treat it as uncitable. Two top-ranking MNQ "ORB backtest" pages **fail arithmetic**: one claims a 314% return with a 4.2% max drawdown at Sharpe 2.50; another a 93.8% win rate with profit factor 2.23 under a stop equal to the full range width. The only credible MNQ number found anywhere is Sharpe ~1.10 out of sample on one contract. Also: the 2025–26 Market Profile web is largely LLM-generated SEO filler that names every day type and never names a threshold, with near-duplicate slugs betraying programmatic generation — and OpenAlex now returns machine-deposited Zenodo DOIs with titles like "New edge proposed: opening_range_breakout_with_prior_day_volume_profile_imbalance".

(b) The most dangerous single number in the sweep: "96.4% of sub-10-point NQ gaps fill." That is an MFE statistic, not a win rate — it says the level was touched before 16:00, not that your stop survived. On the same page sits a 70.7% "continuation rate" beside a **median MFE of 5.50 points against a median MAE of 10.25 points**: the median ORB trade goes **twice as far against you as it goes for you**. Both numbers are true; only the race matters. This is our own MFE trap, arriving from outside.

(c) A bug class to check in any ORB backtest: a commercial backtest platform shipped **look-ahead in its ORB module in July 2025** and a practitioner lost "more than half" of his reported returns to it. The first test of any ORB backtest is a one-bar-forward shift assertion.

6. The honest base rate, and what the profitable minority do differently

Two numbers must never be confused, and nearly every argument about day trading confuses them.

QUESTION	ANSWER	SOURCE
Profitable net of fees in a given year	17-20%	Barber et al., entire Taiwan tape, ~450,000 traders/yr · independently Jordan & Diltz on a US sample: "approximately 20 percent were more than marginally profitable"
Predictably profitable — green again the following year	under 1%	~4,000 of 450,000. "Less than 1%... able to predictably and reliably earn positive abnormal returns net of fees."
Index-futures day traders who persisted past 300 days and lost money	97%	Chague/De-Losso/Giovannetti: every individual who began day trading the Brazilian mini-index future 2013-15, n=19,646. 1.1% beat minimum wage. 0.5% beat a bank teller's starting salary — 8 people, 0.041% of starters.
Mandated broker disclosures	74-89% of accounts lose	FCA 2018 (78%), ESMA
Does experience weed people out?	Almost nobody	Unprofitable Taiwanese traders with 50+ days' experience have a 95.3% chance of trading again within 12 months, against 96.4% for profitable ones. A forum full of confident veterans is exactly what this data predicts.

Now the other half, at full weight, because this is not a "nobody wins" literature.

- **Skill is real, persistent and measurable.** Taiwan's top 500 went on to earn +61.3 bps/day gross and +37.9 net, against −11.5/−28.9 for prior losers — a **73 bps/day spread the authors call "enormous"** next to the 4-5 bps typical of retail-skill studies. Repeat-winning occurs at 1.7-2.0× the chance rate ($p < 0.01$). Confirmed out of sample in a futures brokerage's own book (Subrahmanyam et al., J. Finance 2024).
- ★ **The strongest pro-case source in the whole sweep is Coates & Page (PLoS ONE 2009):** 53 professional futures day traders on the London floor trading their own capital. Mean monthly Sharpe **0.70**. **First-year traders −0.04. Two years or less: 0.39 (not significantly different from the index, $p=0.41$). More than two years: 1.02, $p=0.0001$ — annualised ~3.53.** And critically, individual Sharpes rose within a fixed 20-month window — **that is learning, not survivorship.** Their central finding: P&L is **linear** in risk taken, and a regression of P&L on (SD of P&L, years of experience, a biological marker) gives **$R^2 = 0.94$** . **Sharpe is bought with years; P&L is then bought with size; nobody in that dataset shortcut the first.**

What the winners do differently, ranked by strength of evidence

#	WHAT	EVIDENCE	DO WE DO IT?
1	They show up every single session.	The most replicated marker in the literature. Bombay (NBER 2023): the only profitable cohort of individual day traders is the one with ActiveRatio 0.90–1.00 — every other attendance cohort loses. Taiwan independently: profitable traders with >40 day-trading days in the last year clear their costs and those with fewer do not.	YES — a systemd timer every 5 minutes
2	They concentrate.	After past performance, "the most important predictor of future performance is the concentration of trading in a few stocks."	YES — two instruments
3	They hold winners longer and cut losers faster — and even profitable professionals fail at this.	A US prop team that made \$1.4m "could increase trading profits by holding winners longer and selling losers sooner" (Garvey & Murphy, FAJ 2004), and the disposition gap WIDENS in small size and low volatility — i.e. exactly our quiet-tape clip regime.	NO — and we have already measured it on our own tape. TRAIL +\$186, TARGET_200 + \$197, tight target + \$62.69.
4	They size down.	Causal: capping US retail FX leverage cut high-leverage traders' losses by 40% with no change in spreads. Pre-cap, the top leverage quintile ran 30–40 percentage points worse net monthly return, and the average high-leverage trader lost 44% per month (Heimer & Simsek, JFE).	Untested — item #7 and #9 in the queue
5	They trade the morning.	In the one documented consistently-profitable proprietary day trader's P&L regression (firm records, 61 days), the only significant time variable was Morning Session (+RM109.83, t=2.34, p=0.020) — while market direction (p=0.44) and holding time (p=0.22) were insignificant.	YES

What does NOT distinguish them — three popular beliefs killed

BELIEF	WHAT THE DATA SAYS
Experience alone.	Brazil re-ran the winners' averages excluding each trader's first 200 sessions as a learning allowance — the mean got WORSE (–R\$49 to –R\$91), median unchanged. "There does not appear to be learning." Taiwan agrees experience is "economically weak relative to past profits." Practice does not manufacture an edge; it compounds one that already exists.
A good win rate.	Retail forex traders win 62.8% of trades and still lose, because losers are held twice as long as winners (1.1M trades). The exact mirror of our 25%-at-1.26. There is no win rate that rescues a bad payoff and no payoff that rescues a bad win rate.
A good signal.	Retail order imbalance positively predicts returns while retail traders lose money (Barber/Lin/Odean, JFQA 2023). We already own the local version of that signal.

★ How long it takes to know — the single most important time constant in the report

The 127 genuinely-profitable Brazilian day traders ran daily Sharpes of **0.119-0.162** (annualised 1.9-2.6 — genuinely good). Since $t \approx \text{Sharpe} \times \sqrt{N}$:

CONFIDENCE	SESSIONS NEEDED	CALENDAR TIME
$t = 1$	38-71	2-3.5 months
$t = 2$	152-282	7-14 months
$t = 3$	343-636	17-30 months

A green month is noise. A red month is noise. That is why forum screenshots carry no information, and why people quit on noise — 78% of forex quitters had a losing week immediately prior (88% of the newest), an $N=5$ decision that needs $N \geq 150$. It is also the strongest argument for judging our own work on mechanism and fill quality rather than on short-window P&L.

Three reasons an honest screenshot still misleads

1. **The day is the wrong unit.** The posters with the longest continuity are explicit that they lose money on most days. "Dustin" on Elite Trader, asked directly whether he loses on 3 of 5 days: "Yes, that is not uncommon for me... I'm right less than half the time but still make a good living." Another: "I had one day this year I made 90k, then bled 6k daily for the next 20 days." **\$1,000 a DAY is a different and much harder object than \$250k a year, and nobody credible describes it.**
2. **The screenshot is always of the trend day**, and trend days are 8-10% of sessions. "vovoperador": "it's only about 10% of days that will trend from the open to the close... If you just manage to avoid getting trapped against a strong trend day, you're likely to make money or take just a small loss on all days."
3. **Realised P&L is selection on the outcome.** Linnainmaa's complete Finnish census: "their realized returns from day trades are high, but these returns are not representative of overall performance because of their strong reluctance to realize losses." A trader with the disposition effect has a closed-trade record that looks excellent and an account that does not, because the losers are still open. **Every "I made \$1,000 today" post is a realised figure.** That is our own MFE trap in a different costume — and it is worth auditing our own dashboards for the same bias.

7. Gold is a different animal, and two bugs found while checking

MGC is half the mandate and almost none of the literature. There is **no peer-reviewed study of MGC specifically** — all gold microstructure work is on GC, GLD or spot XAU — and there is **no quantified, published, tradeable intraday gold strategy literature at all**, which means our 13 months of MGC 1-minute may be a larger quantified intraday micro-gold sample than anything public. That is an asset and it leaves us no external benchmark.

FINDING	MECHANICS / NUMBER	GRADE
The hard bound on the biggest gold anomaly there is	Caminschi & Heaney (JFM 2014) compute it themselves: the average price variation in the 15 minutes after a London fixing starts is \$4/oz = \$40 per MGC lot . Captured at BOTH fixings every day for 260 days that is ~\$80/day per MGC lot with PERFECT FORESIGHT OF THE DIRECTION AND ZERO COSTS (~\$200/day/lot scaled to \$3,600 gold). The single largest recurring intraday gold anomaly, traded perfectly and for free, needs 5-12 lots to make \$1,000/day. Δ And that was the pre-2015 telephone fix; the leakage channel was legislated away in March 2015 when it became an electronic ICE auction. The volume/range window should survive; the predictability probably does not.	A
Gold's day session drifts DOWN — three independent sources	Blose & Gondhalekar (COMEX front month 1985–2012): "Overnight returns are significantly positive, whereas day returns are significantly negative." Replicated by the same authors in 2018 across COMEX futures, London Fix spot, gold miners, CEFs and ETFs. Donati & Jung (5-min gold 2001–2018): the annualised cumulative return runs 0% $\rightarrow$ +12.3% from 18:00 to 02:00 NY (the Asian session), gives all of it back to +0.9% by 10:00 NY, then recovers to +8.3% by the close. Essentially all of gold's 2001–2018 return was earned overnight. A long bias at 13:30Z, imported from MNQ, points the wrong way in gold.	A
Gold's volatility clock is the OPPOSITE of the index's	Three sources agree. Batten et al. (PLoS ONE, 5-min 2000–2015): gold volatility is "fairly constant up to 12 PM GMT, increases slightly until 2 PM GMT. After this point volatility DECREASES and levels off." Iwatsubo et al.: NY-session volatility "declin[es] approximately linearly." So 13:30–15:00Z is MNQ's volatility PEAK and MGC's volatility DECLINE — one 90-minute habit applied to two instruments with opposite volatility clocks.	A
The honest cost test on gold's session clock	Donati & Jung charged 100% of the quoted spread: four session strategies went from +10.0%/+18.6%/+28.6%/+7.6% gross to –22.0%/–13.5%/–35.5%/–47.7%. In the modern cost regime (2013–18, spread 0.03% of mid, comparable to MGC) they were +6.2%/+1.7%/+7.9%/–9.3%. Best case = \$41/day GROSS per MGC lot, ~\$11/day NET. \$1,000/day off gold's session clock needs ~90 MGC lots.	B
Intraday momentum is an EQUITY-INDEX effect, not a gold one	Baltussen et al., the same 46-year 60+-futures sample: equity index futures $\beta=4.86$ ($t=6.52$), $R^2_{OOS} +2.88\%$, gross Sharpe 1.73. COMMODITY futures $\beta=1.33$ ($t=3.04$), $R^2_{OOS} -0.01\%$ on the first-half-hour signal, gross Sharpe 1.42 on 4.34%/yr = ~\$6/day per MGC lot , and the authors explicitly warn it "might not be exploitable... after accounting for transaction costs", naming only S&P 500 futures as liquid enough.	A
The only \$1,000-sized gold mechanism is an FOMC surprise — and the literature cannot agree on its sign	Scrimgeour (AJAE): "a 10 basis-point surprise increase in interest rates causes commodity prices to fall immediately by approximately 0.6%", metals more than agricultural — at \$3,600 gold that is \$216 per MGC lot per 10bp, ~\$540 for 25bp . Awartani et al. (5-min NYMEX gold, 1.24m observations): a hawkish surprise lowers gold and "the adjustment 10 min after the shock is significantly THREEFOLD higher than after 5 min" — the move extends rather than reverting. But Gospodinov & Jamali find contractionary surprises INCREASE gold. Two published papers, opposite signs. All sources agree the second moment rises — which is our own conclusion in gold's clothing: volatility is forecastable, direction is not.	A
		A

Gold's order book has a convicted history of being faked

CFTC, 29 Sept 2020: JPMorgan paid **\$920.2m**, the largest CFTC monetary relief ever, for "manipulative and deceptive conduct and spoofing that spanned at least eight years and involved hundreds of thousands of spoof orders in precious metals", 2008–2016. The FCA fined Barclays £26,033,500 for a single trader moving the 15:00 gold fix below a \$1,558.96 option barrier to avoid a \$3.9m payment. **Our pre-registered order-book thinning test needs a "was this order ever fillable" control that an index feature may not.** The counterweight matters too: Batten, Lucey & Peat show the run-length "price suppression" screen the class actions relied on is reproduced by volatility clustering alone — a control that fails, in our own house style.

The sub-30-minute sign, and the cheapest high-value test available

Safari & Schmidhuber (arXiv 2501.16772), 14 years of futures tick data across 24 markets including gold: regressing the next-minute return on normalised trend strength gives, for trend horizons up to 16 minutes, **$b = -0.912\%$ ($t = -13.6$) — MEAN REVERSION**; for horizons ≥ 1 hour, **$b = +0.132\%$ ($t = +2.9$) — TREND. The crossover is approximately 30 minutes. Out-of-sample R^2 is 0.42 and 0.009 basis points respectively. This independently reproduces our five-method 0.50 result from outside, explains it (the true predictable component is half a basis point of R^2), and tells us which way the crumb points: **our momentum families operate in the REVERSION band.** $\triangle$ Below 16 minutes the coefficients are NOT universal — they differ by asset class and time zone — so measure MNQ's and MGC's own b_T rather than inherit a pooled number. **Needs no new data.****

A

★ **BUG 1 — MGC's opening range forms at the COMEX pit open 08:20 ET (~12:20Z), seventy minutes before the equity cash open we watch.** Any MGC logic anchored on 13:30Z is measuring mid-session as though it were the open. This is a bug class, not a research idea.

★ **BUG 2 — the contract roll, and it is happening tonight.** MNQU2026 expires Friday 2026-09-18. The measured MNQ roll crossover is the **Sunday Globex reopen of expiry week** — volume went 2.7% → 17.3% → 66.4% across a single Friday-to-Sunday boundary in June — which means **tonight, 2026-09-13**. Be on MNQZ2026 from Sunday night; the Sep→Dec carry is **+297.25 points**, so every level, VWAP anchor and ATR carried across is 297 points stale unless back-adjusted. And **MGC rolls at FIRST NOTICE DAY, about four weeks before expiry, not at expiry**: the Aug-2026 contract's volume left on 2026-07-30, one day before first notice and 28 days before the 08-27 expiry, moving 6.8% → 35.1% → 88.8% in two days. **A single date-arithmetic roll rule for both instruments is wrong and would leave us in a 0.5%-of-volume contract for a month.** Next MGC roll ~2026-09-29. Also unresolved: GC's front month is GCZ2026 (Dec) while MGC's nearest expiry is MGCV2026 (Oct) — nearest-expiry and most-liquid are different contracts for gold, so confirm which MGC contract we quote before trusting any MGC ATR.

★ **BUG 3 — a one-line calendar fix worth having.** The best-evidenced intraday effect in the entire literature is Baltussen's r_{ROD} (previous close → 30 minutes before today's close, predicting the last 30 minutes: $R^2_{\text{OOS}} +2.88\%$, $t=7.29$, gross Sharpe 1.73, and the authors state explicitly it yields a positive net Sharpe at one tick of cost $\approx$ MNQ's friction). That window sits at 19:30–20:00Z under EDT — fine — but at **20:30–21:00Z under EST, where our 20:40Z flatten cuts it in half.** "Never hold overnight" is untouched by moving the winter flatten to 20:55Z. Nothing in STATE.md notices this. My arithmetic on

their gross number: **≈\$10.57/day net on one MNQ lot** — which is the honest academic figure to print beside \$1,000/day.

★ **THE COST REFRAME, which I think is the single most useful thing in the report after the \$1,000 arithmetic.** MNQ's 1.25 points all-in is **0.0052% of notional**; MGC's 0.45 points is 0.0125%. Holmberg et al. (Finance Research Letters 2013) called **0.08% round trip** "relatively small" for crude oil futures, in a peer-reviewed paper. **Our friction is roughly 15× cheaper in relative terms than a cost a published futures paper waved away.** It is 4.0% of our own \$62.69 target, 100% of a scalp, and 0.5% of a typical session range.

WE DO NOT HAVE A COST PROBLEM. WE HAVE A HOLDING-PERIOD PROBLEM.

And four unrelated sources agree on the lever. Boyarchenko/Larsen/Whelan measured the overnight drift at pre-cost Sharpe 1.1 and **post-spread −0.5** at a one-hour hold — and **widening the window from 1 hour to 2 hours took net Sharpe from −0.5 to +0.3 with NO change to the signal.** Their explanation is the sentence we need: market makers "position their limit order books to incentivize trades that bring their inventory closer to their targets, and do so by MAKING THE TRADE, WHICH PAYS THE BID-ASK SPREAD, NON-PROFITABLE." **An edge the size of the friction is the EQUILIBRIUM, not bad luck** — which is precisely what a 1.05–1.50pt ceiling against 1.25pt of friction is. Add Zarattini's grid, Wu et al. on profit targets, and our own two profitable exit buckets both being the two that held. **The door is one lever wide, and the lever is how long a winner is allowed to run — not which way price goes next, which is the only thing we have already proven, five ways, we cannot know.**

And a warning about decay, from the best-documented anomaly in our own contract family

The overnight drift was a **22-year, $t=7.1$, bootstrap-verified, mechanism-identified, RFS-published** effect: the 02:00–03:00 ET hour returned 3.7%/yr and was **more than 60% of the ES contract's entire close-to-close return**, causally pinned by a daylight-saving natural experiment. Ten weeks ago, on 1 July 2026, the original authors published "**The Disappearing Overnight Drift**": that window has averaged close to zero since 2021, **"the same pattern holds in the E-mini Nasdaq-100 (NQ)"**, and the channel that died was **end-of-day order-imbalance dispersion, halving from 6.5% to 2.9%** because algorithmic liquidity providers slice flow more finely. Two ETFs launched to harvest it closed after fourteen months.

★ Two consequences. **(a)** Ship a **decay monitor** with every edge: name the mechanism, name the observable channels that drive it, and monitor those — P&L is far too noisy to detect decay in time. **(b)** The specific thing that died is order-flow imbalance dispersion, which is exactly the quantity our 250ms depth capture is mining. The Fed has just documented it being compressed out of the market by better execution algorithms. Related: the best practitioner strategy in the sweep has been independently replicated and independently found to have decayed — Paz (2026) reproduced the Noise Area in-sample at Sharpe 1.34 against the published 1.33, then got **0.39 out of sample** over May 2024–Mar 2026, and Zarattini's own live tracking table agrees (2025 YTD +1.0%, five of six months negative). The honest statement is that the forward Sharpe is **unresolved** — that strategy also had 2016 at −12.8% and 2017 at −6.9% back-to-back before +61.1% in 2018 — but **anyone sizing off 1.33 is sizing off a number that has been untrue for two years.**

8. What I would do Monday

Ranked, concrete, shadow-first. Nothing here touches the live order path, and the first six items need no new data and no new signal. Every one is a large-gap comparison, per framing rule R1.

#	DO THIS	BECAUSE	COST / RISK
1	Plot the MAE distribution of the 879 live trades, split by eventual outcome. Winners' MAE on one curve, losers' on another. Read off the percentile that keeps 90–95% of eventual winners alive.	It is the one measurement that settles whether 400 stops against 136 targets is a stop problem or a target problem, and we have been arguing that question for months without it. Three lenses nominated it independently and it is nearly absent from academia too.	One afternoon. No new data. Δ Compute the race, not "reached N R". Δ Report as a lower bound (R4).
2	Plot the daily loss distribution. Is the worst day a large multiple of the median losing day?	If losses are single-session-dominated, a hard daily loss limit is the highest-value change available and needs no edge at all. If it is a slow bleed, the limit buys nothing — and that itself is the finding: our failure mode is not the industry's. One query, never run.	Twenty minutes.
3	Build RelVol and re-score everything. RelVol(w) = first-w-minutes volume $\div$ trailing 14-session same-window mean, $w \in \{5,15,30,60\}$, roll-aware. Bucket the 879 live trades and all 35 shadow arms into deciles. Add NR4 / NR7 / Triangle / prior-trend-day flags off daily bars and a VXN bucket.	The largest gap in the desk — we have never built a selection filter of any kind — and the only finding in the sweep with a published dose-response ($-0.02R$ / $+0.08R$ / $+0.38R$). Four traditions 34 years apart converge on it. It is a gate, not a trade: it costs nothing to run and reduces friction by cutting trade count.	One afternoon. Pure re-scoring of data on disk.
4	Rank the 879 trades by ordinal position within the day and find the trade number at which expectancy goes negative.	One GROUP BY, never asked. If it crosses early we get a free daily trade cap — and the whole one-trade-a-day discourse in the forums is a human discretion-control device whose mechanical analogue is churn, which we have already measured in ourselves.	Ten minutes.
5	Run the 2$\times$2 stop$\times$target factorial and judge it on skew. (a) stop only, no target; (b) target only; (c) both; (d) neither, exit at 20:40Z. Pre-register the prediction that (a) has a lower hit rate and a much higher payoff than (c).	Four quantified sources plus a published theorem plus our own two best exit buckets all point the same way, and this is the resolvable four-cell version of the 430-cell sweep that could never have resolved anything. Write into the badcall ledger first: a falling hit rate is not evidence of a worse exit.	A day. Δ Judge on payoff, skew and worst-day, not hit rate.
6	Test whether flow-since-entry separates recoverable stop-outs from unrecoverable ones. For every one of the 400 stop-outs, compute signed aggressive volume since entry against price displacement	Route A of the loss-pool arithmetic. Cutting the average loss 57% is break-even with no other change, and the vendor's claimed 50% closes 88% of the gap. State 5 — flow your way, price not moving — is the cell a naive sign(OFI)==sign(position) rule gets backwards, and our	Two days. 250ms L2 already on disk. Pre-register a random-early-exit control.

	since entry, both meters reset at entry.	shelved contemporaneous OFI is the perfect instrument to see it.	
7	Re-run Mesfin's 14 MNQ signal families at 1.25 points instead of the 2.00 he charged , on 10.5 years of NQ rather than his 4 years of MNQ. And separately, re-test gap-continuation short (t=3.23, n=22) on ~2.5× the span.	The 0.75-point gap is 60% of our friction and sits in the middle of his own 0.07–1.50 point ceiling. A family at 1.4 points gross is dead at 2.00 and alive at 1.25. And the gap short is the only MNQ result in the whole sweep clearing t>3.0 — PARKED, NOT REFUTED , and our own standing rule says never kill a lead that has a glimmer.	Two days. Existing harness.
8	Point OFI at the FILL. On every existing entry in the record and the shadow book, delay by one adverse micro-bar with an urgency timeout, and measure the change in average fill price in ticks .	The only idea in the sweep that does not require predicting direction. Our OFI profile (r=+0.688 now, −0.001 at 60min) is the textbook definition of a fast alpha, and our surviving ofi-veto-grind cell (−\$172→+\$772) is already this mechanism. Fill quality resolves in dozens of trades; P&L needs thousands.	Two days. Measure in ticks, never in P&L.
9	Replicate the Kaufman volatility breakout exactly as published on 10.5y NQ and 13mo MGC: bands = session open ± 0.5×ATR(14) daily, stop = return to the session open, no target, flat at 20:40Z, decisions only on the quarter hour, 1.25pt / 0.45pt charged.	The cleanest replication target in the sweep: every parameter is given and taken from a 2013 textbook rather than fitted, the stop is one fixed level so there is no path-dependency to model, it has a 19-year reference (CAGR >13%, Sharpe 0.87 net), and our flat-by-20:40Z rule is its EOD rule. △ Check MGC's 0.5×ATR14 band against the gate entry floors first.	Two days. Shadow only.
10	Settle the published disagreement on our own contract. Build BOTH bands on the same NQ tape — Seeck's fixed daily VXN/16 band from the prior close, and Zarattini's time-of-day-varying band from today's open — tag every breach with (band type, time of day, VXN bucket, breach size in ATR) and measure the return path from +5 minutes out to the flatten.	Zarattini on SPY says a band break TRENDS to the close (Sharpe 1.33). Seeck on NQ says it REVERTS — 85.2% same-session reversion on 860 breaches, and he explicitly rejects the momentum version on the same data, with out-of-sample Sharpe 1.29 better than in-sample 0.47. These are probably not contradictory — they are different horizons, different instruments and, crucially, NOT THE SAME EVENT (a breach of a fixed daily band at 10:00 is far more extreme than a breach of a narrow time-of-day band at 10:00). One run answers it, and Seeck is on the right instrument.	Two days. This also bears directly on our own "confirmed chop → bench all momentum, keep reversion" rule.
11	Measure $\lambda \approx 1/\text{depth}$ as a LEADING volatility estimate and test it as incremental R ² over lagged ATR.	The one place a C-grade practitioner claim and an A-grade replicated identity converge on a mechanism. Every stop width, position size, expected exit slippage and target	Two days. 250ms L2 on disk.

feasibility number on this desk currently comes from trailing ATR; λ is the leading version of the same quantity **and it never asks the book which way price will go.**

12 Fix the three bugs, and buy nothing yet. (a) Re-anchor MGC's opening range to 12:20Z. (b) Roll to MNQZ2026 and back-adjust the +297.25 carry; write a per-instrument roll rule (MNQ at expiry-week Globex reopen, MGC at first notice day). (c) Move the winter flatten to 20:55Z so the r_{ROD} window is not cut in half. (d) Price the Sierra + Denali data upgrade at ~\$57/month and diary it for go-live, not now — non-pro exchange rates require a funded account.	Three of these are bug classes that quietly corrupt measurements we already trust; the fourth closes the one real data gap in the desk, and nobody had priced it.	Half a day. $\triangle$ Warn before restarting anything.
13 Adopt the statistics layer, and place ourselves on the allocator's table. Deflated Sharpe with N = the honest hundreds of configurations tried; Minimum Track Record Length on the live book; PBO/CSCV instead of a single hold-out; the $t > 3.0$ bar. And tonight: compute annualised return and vol for the live book and every shadow arm.	We are a multiple-testing machine that ranks on raw P&L. FundSeeder scores a 12.3%-return/5.7%-vol team at 87.1 and a 470%-return/86%-vol trader at 62.8 — consistency beats magnitude about 5:1 and a 17-year record scores 61, so length buys nothing. MinTRL turns "are we any good?" into "we need N more trades before we could tell."	One day.
14 Add breadth as shadow sleeves — MES, MYM, M2K, MCL, M6E. Not more gold.	Carver's measured ladder: 1 instrument SR 0.35 $\rightarrow$ 8 at 0.61. +74% Sharpe with no new signal, and his explicit \$30,000 portfolio is five instruments — written before the micro complex dissolved the problem he was solving. $\triangle$ The free half applies today: at max position 1 lot his own prescription is a threshold filter, the same answer as item #3.	The slowest item — needs new sleeves. Largest number on the page.
15 ★ Measure your own blinded edge. <code>r-candles.com</code> is free and is the same blinded, bias-free replay apparatus used in the anonymised-chart experiment (no ticker, no date, survivorship-free, 30+ years, 40,000+ stocks).	It is US stocks and swing-oriented, so it measures chart-reading skill generally and not MNQ intraday. But given our own banked findings — the operator's hands are the exit (3/3 manual claims = \$726 in 23 minutes, no fixed level reproduces it) and claiming cannot be backtested — measuring your personal edge in a blinded harness is, in my judgement, the highest-	Free. You can start tonight.

value BUILD suggested by tonight's reading, because it calibrates whether it is worth constructing the MNQ version off our 10.5 years of NQ 1-minute.

And one thing NOT to do. Do not relax the never-hold-overnight rule. A 20-year SPY study gives pure-intraday Sharpe **0.88** against 0.75 for carrying overnight, with drawdown nearly doubling (−25.6% vs −45.4%). **Your hardest rule is correct, and its cost is about 1% of CAGR, not the fortune the forums imply.** (One honest split to note: overnight helps the long sleeve 0.76→0.97 and destroys the short sleeve 0.52→0.17.) Likewise do not port VWAP Trend Trading to MNQ — its Sharpe 2.1 rests on per-share commissions, and at ~15 round trips a day MNQ's fixed \$2.50 is \$37.50/day/lot. And do not quote Sharpe 2.81 as an MNQ number: it needs a 20-wide opportunity set where 3.1% of days carry 50% of the trades, and we have two instruments.

What the composite realistically pays. Assemble everything that converges — the window, the RelVol day gate, a veto list, a dwell-confirmed entry at structure, a hybrid stop, two tiers plus a scratch, an MAE-conditional time stop, no fixed target, partials no earlier than 2R, never to breakeven, a trail on max(band, VWAP), size frozen at the session open, layered rails, and the Rule of Three — and you get roughly **1-2 trades a day at a 20-40% hit rate with a 3R+ tail, standing down on most sessions.** Priced honestly at \$30k that is **\$35-\$140/day central and \$100-\$300/day optimistic.** Which is not \$1,000 — but it is also not nothing, it is the top of the verified distribution, and **every single component of it is testable on data already sitting on this disk.**

9. Sources

Every claim in this report traces to one of these plus the lens NOTES.md files. Grades as in §5. [△] Marked items are the ones a human should verify by hand: SSRN and several publishers block automated fetches from this box, so those rest on indexed abstracts rather than full text.

WORK	WHAT IT GIVES	URL
Baltussen, Da, Lammers & Martens, JFE 142(1) 2021 — 60+ futures, tick data, 1974–2020	The best-evidenced intraday effect there is. r_{ROD} R^2_{OOS} +2.88%, $t=7.29$, gross Sharpe 1.73; the two-horizon agreement filter; net-positive at one tick of cost. Also the dealer-gamma mechanism.	pure.eur.nl/ws/files/58145484/
Gao, Han, Li & Zhou, JFE 129(2) 2018 — Market Intraday Momentum	The paper that created the literature: first half-hour predicts last half-hour, $R^2=1.6\%$. Fails out of sample in the US twice.	doi.org/10.1016/j.jfineco.2018.05.009
Rosa, J. Futures Markets 42(12) 2022	Failed replication #1: "the predictability disappears in the out-of-sample period"; a threshold version survives.	doi.org/10.1002/fut.22375
Li, Sakkas & Urquhart, J. Financial Markets 57 2022 — 16 markets	Failed replication #2: US in-sample R^2 3.41% but $R^2_{\text{OOS}} = -3.10\%$, the worst of all sixteen ; expansion R^2 only 0.84% vs crisis 7.53%. Charges zero costs anywhere.	centaur.reading.ac.uk/95566/1/
Boyarchenko, Larsen & Whelan — NY Fed SR917 → RFS 36(9) 2023	The overnight drift: 3.7%/yr, $t=7.1$, >60% of ES's close-to-close return, DST natural experiment — and pre-cost Sharpe 1.1 → post-spread -0.5 , rescued to +0.3 by doubling the hold.	newyorkfed.org/.../sr917.pdf
Boyarchenko, Larsen & Whelan — Liberty Street Economics, 1 July 2026	"The Disappearing Overnight Drift": zero since 2021, "the same pattern holds in the E-mini Nasdaq-100", imbalance dispersion 6.5%→2.9%. The decay case study.	libertystreeteconomics.newyorkfed.org/2026/07/
Kaminski & Lo, J. Financial Markets 18 2014	The stop-loss theorem: under a random walk a stop always reduces expected return; it adds value only with momentum.	doi.org/10.1016/j.finmar.2013.07.001
Han, Zhou & Zhu — "Taming Momentum Crashes", 88 years	A 10% stop cut momentum's worst month -49.8% → -11.4% and more than doubled the Sharpe — all in skew and worst-day, not the mean.	doi.org/10.2139/ssrn.2407199
Tsai, Wu, Syu, Lei, Wu & Ho, IEEE Access 7 2019 — TORB	Best opening-range probe on E-mini Nasdaq is one minute (17.51%/yr); the cash-hours alignment does the work, not the breakout (TRB 5.58%). Δ Probe time searched on the full sample.	ieeexplore.ieee.org/.../08641124.pdf
Holmberg, Lönnbark & Lundström, Finance Research Letters 2013	"ORB is basically LONG VOLATILITY"; the result is not robust across sub-periods; and the 0.08% round-trip cost they call "relatively small" — the anchor for our cost reframe.	dx.doi.org/10.1016/j.finmar.2013.05.006
Cont, Kukanov & Stoikov, arXiv: 1011.6402	Δ price is linear in OFI with slope inversely proportional to depth — OFI is a contemporaneous identity, and $\lambda \approx 1/\text{depth}$.	arxiv.org/abs/1011.6402
Takahashi, arXiv:2508.06788 — S&P E-mini, 1-second SVAR	"Shocks dissipate almost entirely within a second." Around news, price impact rises while flow volatility falls — i.e. depth collapsed.	arxiv.org/abs/2508.06788
Safari & Schmidhuber, arXiv: 2501.16772 — 14 years, 24 markets incl. gold	$b = -0.912\%$ ($t=-13.6$) up to 16 min (reversion); +0.132% ($t=+2.9$) at ≥ 1 hour (trend); crossover ~30 minutes . R^2_{OOS} 0.42 and 0.009 basis points.	arxiv.org/abs/2501.16772
Maitrier & Bouchaud, arXiv: 2506.07711	Volatility is largely the mechanical result of uninformed metaorder flow — predicts direction unforecastable and volatility forecastable simultaneously.	arxiv.org/abs/2506.07711
Bailey & Lopez de Prado — Deflated Sharpe, PSR / MinTRL, PBO-CSCV	The statistics layer we do not use. Companion bar: Harvey, Liu & Zhu (RFS 2016) — $t>3.0$, not 2.0 .	doi.org/10.2139/ssrn.2460551
Δ Mesfin 2026, SSRN 6709401 — MNQ-specific , 947 RTH days, 14 signal families	Gross-edge ceiling 0.07–1.50 points , nothing survives; charged 2.00pt round trip. Glimmer: gap-continuation short $t=3.23$, +14.52pt mean net, $n=22$.	doi.org/10.2139/ssrn.6709401

△ Mesfin 2026, SSRN 6792161 — MNQ ML null, 944 days	Base rate 51.8%; gradient boosting 50.00–50.89%, LSTM 50.59% — every model worse than always guessing UP . This is the answer to "should we throw AI at it".	doi.org/10.2139/ssrn.6792161
△ Seeck 2026, SSRN 7364204 — NQ-specific , 860 breaches 2018–2026	Band = prior close ± VXN/16; 85.2% same-session reversion , exit 30 min, OOS Sharpe 1.29 > IS 0.47, and the momentum version is explicitly rejected on the same data.	doi.org/10.2139/ssrn.7364204

Academic — gold and commodities

WORK	WHAT IT GIVES	URL
Caminschi & Heaney, J. Futures Markets 2014 — GC futures 2007–2012	The fix window: +47.8% volume and +32.9% one-minute volatility in the first minute, elevated for thirteen minutes, $t > 10$; direction predictable ~70% overall and "above 95%" for large fixes. And the \$4/oz bound that caps the whole anomaly at ~\$80/day per MGC lot with perfect foresight.	doi.org/10.1002/fut.21636
Blose & Gondhalekar, Applied Economics Letters 2014; and with Kort, J. Economics & Finance 42(3) 2018	"Overnight returns are significantly positive, whereas day returns are significantly negative" — replicated across futures, spot, miners, CEFs and ETFs.	doi.org/10.1007/s12197-017-9403-0
Batten, Lucey, McGroarty, Peat & Urquhart, PLoS ONE 2017 — 5-min gold 2000–2015	Gold volatility is flat to 12:00 GMT, peaks ~14:00 GMT, then decreases — the opposite of the index clock.	doi.org/10.1371/journal.pone.0174232
Hauptfleisch, Putniņš & Lucey, JFM 2016 — 17 years	After the 2007 COMEX electronic upgrade "there are no longer clear intraday patterns... Daylight or working hours no longer affect the location of price discovery." Any gold session rule from an old sample must be re-derived.	doi.org/10.1002/fut.21775
Awartani, Hussain & Virk, IRFA 2024 — 5-min NYMEX gold, 1.24m obs	Hawkish surprise lowers gold; "the adjustment 10 min after the shock is significantly THREEFOLD higher than after 5 min". Opposed in sign by Gospodinov & Jamali.	doi.org/10.1016/j.irfa.2024.103486
Scrimgeour, AJAE 2015	A 10bp rate surprise moves commodities ~0.6%, metals most — ~\$216/MGC lot per 10bp .	doi.org/10.1093/ajae/aau054
Batten, Lucey & Peat, Economic Modelling 2016	The gold "price suppression" run-length screen is reproduced by volatility clustering alone — the control that fails.	doi.org/10.1016/j.econmod.2016.03.005
CFTC, 29 Sept 2020 — JPMorgan, \$920.2m	Eight years, hundreds of thousands of spoof orders in precious metals futures. Plus the FCA's £26,033,500 Barclays gold-fix fine.	cftc.gov/PressRoom/PressReleases/8260-20 · fca.org.uk/.../barclays-fined
ICE Benchmark Administration — the LBMA Gold Price	10:30 and 15:00 London, 30-second rounds. 15:00 London is 10:00 ET all year, so the PM auction lands on the US 10:00 ET data slot.	ice.com/iba/lbma-gold-price

Base rates, skill and track records

WORK	WHAT IT GIVES	URL
Barber, Lee, Liu & Odean, J. Financial Markets 18 2014 — the entire Taiwan Stock Exchange, 1992–2006, 2.34M trader-years	17–20% profitable in a year; under 1% predictably so ; top 500 go on to earn +37.9 bps/day net = \$114/day on \$30k ; top-vs-bottom spread >70 bps/day. Best predictors: past Sharpe, then concentration .	escholarship.org/content/qt7k75v0qx/
Chague, De-Losso & Giovannetti — EVERY individual who began day trading Brazilian mini-index futures 2013–15, n=19,646	The closest-matched population anywhere. 97% of those who persisted past 300 days lost money. 1.1% beat minimum wage. 0.5% beat a bank teller's starting salary. Equities companion: of 98,378 starters, 99.43% never reached 300 sessions; of the 554 who did, mean daily gross –R\$49.	doi.org/10.2139/ssrn.3423101
Coates & Page, PLoS ONE 2009 — 53 professional London floor futures day traders, own capital	The strongest pro-case source. Mean monthly Sharpe 0.70; year 1 = –0.04, ≤2yr = 0.39 (p=0.41), >2yr = 1.02 (p=0.0001, annualised ~3.5) , rising within individuals. P&L is linear in risk taken; R ² =0.94.	journals.plos.org/plosone/.../0008036
Linnainmaa — "The Anatomy of Day Traders", complete Finnish census	"Their realized returns from day trades are high, but these returns are not representative of overall performance because of their strong reluctance to realize losses." Why an honest screenshot still misleads.	doi.org/10.2139/ssrn.472182
Dalvi, Deuskar, Glosten & Jagannathan, NBER 2023 — Bombay	The only profitable individual cohort is ActiveRatio 0.90–1.00 . Individual day traders lose 3.2bp = 73% of the half-spread.	nber.org/... (NBER series)
Ben-David, Birru & Prokopenya, NBER w22146 — full retail FX broker records, 1.1M trades	Trade-size discontinuity at exactly zero past-week return worth 12–17% of an SD, with no corresponding discontinuity in future performance. Traders win 62.8% of trades and lose money.	nber.org/papers/w22146.pdf
Heimer & Simsek, JFE — US retail FX leverage cap	Causal: capping leverage cut high-leverage traders' losses 40% with no change in spreads; the average high-leverage trader lost 44% per month .	doi.org/10.3386/w18706 (related series)
Garvey & Murphy, Financial Analysts Journal 2004	A US prop team that made \$1.4m "could increase trading profits by holding winners longer and selling losers sooner"; the gap widens in small size and low volatility.	doi.org/10.2469/faj.v69.n4.1 (journal)
FundSeeder — Jack Schwager's platform, records verified by read-only broker connection	"20K traders worldwide", 15 seeded = 0.075% . The four index-futures records at 12.3% / 22.5% / 27.2% / 55.5%, and the FS Score that ranks a 12.3%/5.7%-vol team at 87.1 above a 470%/86%-vol trader at 62.8.	fundseeder.com/our-traders
World Cup Advisor — live, commission-net, publicly-tracked accounts	Pau Perdices : discretionary intraday MGC+MNQ+MES, \$10,000, 8.31 months, +\$7,163 net through –21.2% ≈ \$124/day scaled to \$30k. Sambataro: +91.6%/+60.5%/+50.3% then –32.8% YTD , max DD –37.4%.	worldcupadvisor.com
World Cup Trading Championships — the full official 42-year table	135 first places, 84 distinct winners, 59 of them (70%) never win again . \$3,999 entry, \$10,000 minimum, entrants may trade more than one account. In the day-trading/micro divisions the worst top-five finishes are 0.0%, 0.3%, 0.4%, 4.7%; one monthly micro division was won with 0.31% and three had no entrant at all.	worldcupchampionships.com/.../historical-standings
AutumnGold — 191 CTA programs; NFA BASIC	The best lifetime compound is ~50%/yr = \$60/day on \$30k; the best single months belong to programs whose lifetime returns are 7.9%, 12.6%, –1.2%, 3.7%, 18.8% against worst drawdowns of –88%, –93%, –66%, –53%, –61%. Comhla "Artemis 3X" (ES/MES/NQ/MNQ, registered CTA) has a negative lifetime compound return at –66.4% DD .	autumngold.com/Advisor/ranking_report.php · nfa.futures.org/basicnet
QTS Capital — Tail Reaper (Ernest Chan's firm, intraday ES); Collective2 C2Star	QTS's published objective: "daily stop loss ≤ 5% of nominal"; actual –23.0% then –29.6%, max DD –67.6%. C2Star certifies on drawdown (5% open loss, 5% 24-hour, ~10% all-time) while the return bar is merely "beat the S&P by +1% over rolling 60 days".	qtscm.com · collective2.com/c2star

Kevin Davey — 2006 World Cup futures champion	On four consecutive 100%+ years: "Monte Carlo simulation told me I had a 50% chance of doubling my money in a year, but also a 20% [chance] of losing two thirds of my starting equity." And his winning system was trend following updated at night , not intraday.	kjjtrading.com/100PercentReturns.pdf
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Quantified practitioner research — the Concretum / Zarattini tier

All Grade B. Survivorship-free samples, commissions charged, methods fully specified — and two of three authors sell day-trading education, which is a real conflict and must be stated. Index of papers: concretumgroup.com/papers

PAPER	THE NUMBERS THAT MATTER	URL
A Profitable Day Trading Strategy For The U.S. Equity Market (Zarattini, Barbon, Aziz) — ~7,000 stocks, survivorship-free, 2016–2023	The selection result. Identical entry/stop/exit: all stocks = total 29%, Sharpe 0.48 (loses to buy-and-hold); RelVol $\geq$ 100% top-20 = total 1,637%, Sharpe 2.81 , alpha 35.8%, beta 0.00. Dose-response $-0.02R$ / $+0.08R$ / $+0.38R$. Opening-range length: 5m 2.81, 15m 1.43, 30m 0.21, 60m 0.40. Every one of the 25 best names wins 17–24% of the time.	.../A-Profitable-Day-Trading-Strategy....pdf
Can Day Trading Really Be Profitable? (Zarattini, Aziz) — QQQ/TQQQ 2016–2023, 1,795 trades	24% win rate, +0.13R/trade, Sharpe 1.13–1.19, alpha 33–48%. And the 2-D stopxtarget grid: the optimum is the tightest stop tested with NO profit target , held to the close — they then honestly undercut their own 9,350% cell on slippage grounds.	.../Can-Day-Trading-Really-Be-Profitable.pdf
Beat the Market (Zarattini, Aziz, Barbon) — SPY 2007–2024, 2,620 days, 7,668 trades, slippage measured from 1,000 live IBKR orders	The Noise Area. Base Sharpe 0.61 $\rightarrow$ max(band,VWAP) trail 1.24 (hit 54% $\rightarrow$ 43%, skew $-1.24\rightarrow+1.29$) $\rightarrow$ vol-targeted sizing 1,985% total, Sharpe 1.33 on the identical trades. Day-selection table: NR4 22bps t=5.14 Sharpe 3.2, NR7 t=3.07, Triangle t=3.19, Inside t=0.85 NULL, Outside t=1.05 NULL, prior trend day –2bps t=–0.24 . Wednesday 18bps t=3.42. RSI5 $\beta=-3.25$, p=0.001.	.../Beat-the-Market.pdf
The Power Of Price Action Reading (Zarattini, Stamatoudis) — blind anonymised charts, 9,794 gap events, 1,580 trades	18% win rate, avg +10.10R against avg –1.02R = +1.03R/trade, Sharpe 1.70. Co-author is a verified 2023 US Investing Championship top performer. The apparatus (r-candles.com) is free.	.../The-Power-Of-Price-Action-Reading.pdf
Improving Performance with Fast Alphas (Zarattini, Pagani, Feb 2026) — SPY 5-min, 19 years	A 5-min reversion signal that is negative standalone after commissions lifts a trend model's CAGR ~200bps and Sharpe 0.87$\rightarrow$0.99 net, with zero added round trips , purely by timing the fill. Also the Kaufman volatility-breakout baseline: CAGR >13%, Sharpe 0.87, no regime filter.	.../Improving-Performance-with-Fast-Alphas....pdf
△ Paz 2026, SSRN 7290621 — the only independent replication in this literature	In-sample reproduces at Sharpe 1.34 vs the published 1.33 — genuinely reproducible, which is rare. Out of sample May 2024–Mar 2026: total 9.4% vs 29.5% buy-and-hold, Sharpe 0.39 . Candidate cause named: 0-DTE proliferation.	doi.org/10.2139/ssrn.7290621 · github.com/PazSheimy/spy-intraday-momentum-oos

Practitioner schools — all Grade C unless noted

SCHOOL / PERSON	WHAT THEY CONTRIBUTE	URL
Al Brooks	The 90-minute claim (~90% of days set an extreme in the first 90 minutes — we measured 90.0% on MNQ, so he is right); H2/L2 second entries; structural targets. His ladder tops out at 4 ES points × 3 contracts ≈ \$600/day starting from \$10/day on micros. △ 13 of 25 manual chapters are members-only.	brookstradingcourse.com/.../day-trading-open
Bob Volman	The Second Break — "arguably no higher probability of a winning trade in the market"; and against breakeven stops: "simply asking for an early exit". His spec translates to MNQ only at a 12-13 point stop and a 25-point target . Publishes six months of day-by-day skip decisions and never states the criterion.	daytradingbias.com/.../bob-volman
Lance Beggs (YTC)	The Rule of Three — the only halt rule in the sweep that is neither P&L- nor direction-based. And the trail rule: "any trailing stop must be at a place where you consider the trade idea is no longer valid. ANY CLOSER IS TOO CLOSE." △ yourtradingcoach.com is now a parked domain ; archive only.	web.archive.org/.../Rule-of-Three.html
John Mack (PATs)	States his own evidence grade: "back testing is bunk... you cannot back test our price action trading strategies." Puts "serious money" at \$500/day . His 4-tick ES scalp translated literally to MNQ is structurally negative before any win rate .	priceactiontradingsystem.com/...
Jigsaw Trading (Peter Davies) — B for the leaderboard	A DOM vendor saying the DOM does not forecast: "fading depth... does not mean the level will hold. Quite often the depth will simply disappear"; both fading depth and fading icebergs are "one rule trading systems"; depth is "around 20% at most" of the picture. The public leaderboard : June 87/282 = 30% profitable, July 105/307 = 34%; winner "Fenix" ~50% hit, \$619/\$489.05 = 1.266:1 ; July winner on a live account made "just under \$5k for the month" ≈ \$250/day, best of 307. And Rule 20 on scale-in mechanics.	jigsawtrading.com/blog/jigsaw-leaderboard-june-and-july-results.../trading-rule-20...
Mike Valtos (Orderflows)	The four stop objects we lack: "most traders use arbitrary stop levels... Wouldn't it make more sense to have a stop level based on what the market is telling you?"; the two-tier mental/hard stop; the MAE-conditional time stop; and the scratch — "You will either get to your profit target or get stopped out. THAT ISN'T THE WAY TO TRADE."	orderflows.com/.../dont-use-arbitrary-stop-levels.../a-tale-of-two-stops
Jim Dalton / Market Profile	The day-type taxonomy and open types, with the thresholds. The best line in the sweep on when not to trade: "No trade is a legitimate trade. When confidence is low you generally want to stay uncommitted so that you can execute if the confidence level changes." △ Never peer-reviewed in 40 years .	jimdaltonttrading.com/glossarypage
Firich (2012), relayed with full tables by ATAS — the only measured Market Profile study	Day-type frequencies: non-trend 6.81%, normal 2.43%, normal variation 22.39%, trend 9.54% , neutral 30.21% — total only 71.38%. P(trend day tomorrow) = 9.50/9.94/10.39% against an unconditional 9.54% . IB broken on 73.7-78.3% of days. △ ATAS then refits the 2× threshold to 1.5× post hoc.	financnik.cz/.../market-profile-statistiky.../atas.net/blog/initial-balance-indicator
Toby Crabel (1990) and Crabel Capital	"Early Entry 2" — first-bar volume against the same bar over the prior 15-20 days: RelVol, 34 years earlier . NR4/NR7 originate here. His own 102-year, 84-market study: portfolio Sharpe "above six" (1970s-90s) → 2.92 (2000s) → 0.91 (2010s) — "It earns much less than it did." And Crabel Capital's stated competitive advantage is "execution costs are among the lowest in the industry."	mebfaber.com/2007/09/17/tony-crabel-and-the-secret-orb.../oxfordstrat.com/trading-strategies/opening-range-breakout
acrary (Elite Trader) — the best method content in the forums	The EDGE TEST : random entries with matched hold lengths, 5,000 Monte Carlo passes, longs and shorts ranked separately per year, 70th percentile on both for 3-5 years, re-check quarterly, stop below 70 ,	elitetrader.com/et/threads/acrarys-edge-test.11220...

and disable stops and targets during the test. Plus the trading-cycle arithmetic and a Monte Carlo VaR table.

[/acrarys-strategy-formation.84891](#)

"calevonlear",
"vovoperador", "Dustin",
"IllusionHD" (forums)

"Trade direction and entry are not as important as sizing and after-entry management... The market is just random. Direction is meaningless, afterwards is what matters." · trend days "only about 10% of days" · "I'm right less than half the time but still make a good living" · the one claim matching the target: **170 MNQ points on 5 lots = \$1,700 by 09:58am.**

[elitetrader.com/.../do-daytraders-profit-everyday... · reddit.com/r/FuturesTrading/.../1je7g30](#)

edgeful — the one quantified vendor tier

NQ base rates on 115–127 sessions. **29.92% of sessions break the 15-minute opening range BOTH ways.** Their 5-min ES ORB takes target = 50% of the range against a stop at the opposite edge, so break-even after our friction is **69–73%** against their measured 72.17%. Δ Their dollar figures are unit-ambiguous (a "NY session ATR of \$449.26" is only coherent at MNQ's \$2/pt) and their landing page claims 75%/83% NQ gap-fill while their own article measures 53%/59%.

[edgeful.com/blog/posts/nq-futures-trading · .../5-minute-opening-range-breakout-es-strategy](#)

Rob Carver — , and the most useful sizing writing found

The breadth ladder (1 → 0.35, 5 → 0.57, 8 → 0.61, 37 → 0.70; discreteness penalty ~20% at one lot, zero at 3–4); his explicit **\$30,000 = five instruments**; the "speed limit" (never spend more than a third of expected pre-cost return on costs, target 10%) which demands **≥3.75 MNQ points of gross edge per round trip**; and operational Kelly, which prices \$1,000/day at **Sharpe 4.10 at half-Kelly**. Also: he built this exact machine in 2025 and posted "Postscript: It didn't work" — simulated Sharpes from +48 to –129 depending only on the stop-fill assumption.

[qoppac.blogspot.com/pot.com/2016/03/diversification-and-small-account-size · .../2025/05/can-i-build-scalping-bot](#)

Prop firms, payouts and the industry's own risk rails — **A** (legally-exposed disclosures)

SOURCE	WHAT IT GIVES	URL
Topstep — "2025 Trader Performance Statistics"	16.8% of Combines initiated pass · 51.8% of individual participants pass at least one · 33.3% of funded participants ever receive a payout · 0.71% ever reach a LIVE funded account. Plus "Over 63% of traders have lost account in a single day." And the marketing that refutes the target: "7,000+ traders paid weekly" × \$5,000/wk × 52 = \$1.82bn/yr against a \$1.4bn FIFTEEN-YEAR all-time total . Their Octagon competition defines a streak day as "more than \$500 in profit".	topstep.com/risk-disclosure · help.topstep.com/.../trading-combine-parameters · .../what-is-the-consistency-target
Tradeify, Earn2Trade, TradeDay disclosures	Tradeify 17.2% / 40.3% / 28.5% / 3.0% to Live , and states the hidden variable: participants "initiated a median of 3 each." Earn2Trade: 8.89% of subscriptions passed, of those 5.23% traded a Live account and 94.77% a LiveSim . TradeDay 36% pass — and TradeDay has no trailing drawdown, the cleanest (if confounded) evidence on which rule kills accounts.	tradeday.com · earn2trade.com
Payout Junction frozen archive — Zenodo doi: 10.5281/zenodo.22342845, CC-BY, 35 months, blockchain-verified settlement transactions only	August 2026 across 39 firms: \$95,535,787 across 62,037 payouts. Mean \$1,540, MEDIAN \$1,125 . 76% under \$2,000; 0.058% at \$25,000+. Over 35 months the total grew 9x and the count 14x while the median sat at ~\$1,200 the whole time — the industry grows by adding small payees. \$1,000/day ≈ \$21,000/month, which that entire pool supports for ~4,549 people on Earth .	zenodo.org/records/22342845
Apex — 4,031 published payout certificates recovered from the Wayback CDX index	Total \$7,539,321, mean \$1,870, median \$2,000, p99 \$3,500, 85.8% ≤\$2,000, four above \$10,000 in the entire corpus . △ The modal values are literally Apex's published first-five-payout maxima, so this measures the cap, not earning power.	apextraderfunding.com (via web.archive.org CDX)
Take Profit Trader — Sustainable Trading Policy	Adversarial, money-at-risk confirmation of our lab-vs-production traps: "The fills are easier on a trader in a simulated environment, so it's actually easier to make profits in the simulated PRO account"; the patterns that do not replicate live include "relying on simulation-specific fill inefficiencies" and "aggressively sized positions at fixed, predictable market moments like the open... or other simulation-favorable windows". A specific warning that a PAPER desk's first-90-minutes record is the least trustworthy part of its tape . Topstep independently bans "repeatedly exploiting the relative lack of slippage in SIM."	takeprofittrader.com · help.topstep.com/.../what-is-responsible-trading
FTMO — 22 paid-trader post-mortems with win rate and RRR from its own broker data	20 complete pairs, bimodal with a hole in the middle : $w \leq 46\%$ always carries RRR 1.58–5.59; $w \geq 55\%$ carries 0.65–1.83. Minimum edge index $w \cdot R / (1 - w) = 1.295$. Not one paid trader sits at $w \approx 25\%$, $R \approx 1.3$ — where we sit, at 0.430.	ftmo.com (WordPress REST API)

Instrument, data and tooling facts

SOURCE	WHAT IT GIVES	URL
Measured on our own tape — MNQ 1-min 239 RTH sessions, MGC 264, top-of-book quotes 12 days each, 10-level L2 at 250ms 8 days each, NQ↔MNQ 82,095 paired minutes, all DST-correct	First-90 = 72.2% of MNQ's RTH range on ~42% of volume · one extreme 90.0% of days, both 10.4%, afternoon extreme 37.2% · MNQ median spread 2 ticks at every hour, worst in London (3 ticks), best 16:00–19:00Z, and 09:30–09:35 ET is the single most expensive five minutes (\$1.36 vs \$0.86) · MGC median 3 ticks, never tightens · 10-lot book-walk \$0.35/lot MNQ vs \$3.60–4.60 MGC · MNQ↔NQ correlation 0.995102 at lag 0 and 0.009 at ±1 min, so neither contract leads · basis >1 tick on 47% of minutes, so the 10.5y NQ series is not a tick-exact MNQ proxy · the roll dates and the +297.25 Sep→Dec carry.	reports/lit_review_2026-09-13/instrument_facts/NOTES.md
IBKR TWS API documentation — the one real data gap	"Streaming market data... is not tick-by-tick but consists of aggregate snapshots taken several times per second", while depth is "sent without sampling nor filtering". A true tick stream exists (reqTickByTickData with AllLast) and we appear not to use it. Also the hard constraint nobody mentioned: depth requests are capped at three at the default 100 market-data lines.	interactivebrokers.github.io/tws-api/md_request.html · .../ tick_data.html · .../ market_depth.html
Jigsaw daytradr V6.1 release notes, 2025-12-05 — the killer corroboration	"Fixed MAE/MFE (not for IB — that's down to a data restriction)". A platform that computes MAE/MFE for Rithmic, CQG, Tradovate, NinjaTrader, StoneX and IQFeed cannot on Interactive Brokers — and MAE/MFE are the primitives behind all 430 exit configurations.	jigsawtrading.com (release notes)
Sierra Chart — packages, Denali feed, .scid format, DTC restrictions	The fix priced: Package 11 (\$46/mo, \$29.90 on 12-month prepay) + Denali CME-with-depth \$13.50 + COMEX \$13.50 non-pro ≈ \$57/month for "tick by tick unfiltered", millisecond stamps, exchange-fact bid/ask trade volume "100% accurate for CME". ⚠ Non-pro rates require a funded account (otherwise \$145/mo per exchange), so this is a go-live purchase. ⚠ DTC cannot serve CME data ("Requests are rejected") — read .scid files off disk instead. ⚠ Sierra's MBO is Package-12-only, ≥3-lot orders only, and not recordable , so it is an eyeball tool, not a research source.	sierrachart.com/.../DenaliExchangeDataFeed · .../ IntradayDataFileFormat · .../ MarketByOrder
IBKR commission schedule and market-data pricing; Databento GLBX.MDP3	Our current data spend is ~\$27.30/month. Databento publishes the latency ceiling: "90th percentile latency up to your application at 42 microseconds (cross-connect) or 590 microseconds (internet)" — i.e. the best retail feed is 100–350× faster than a human. Latency is not our gap.	interactivebrokers.com/...?f=14193 · databento.com/datasets/GLBX.MDP3

What I could NOT get, stated so nobody repeats the dead ends

The session's WebSearch budget (200/200) was exhausted by concurrent agents before most lenses issued a single query, so nearly all of this was done by direct HTTP against OpenAlex, Crossref, Unpaywall, Semantic Scholar, arXiv, RePEc, Zenodo, publisher repositories and the Wayback CDX API, with PDFs saved to disk and read with `pdftotext`. Hard-blocked from this box: **SSRN, ScienceDirect, Wiley, Taylor & Francis, Oxford Academic, Google Scholar (partly), ResearchGate, cmegroup.com (every path, 60s timeouts), iasg.com, live collective2, darwinex, futures.io and elitetrader direct, reddit's JSON API**, and every general search engine (Google/Brave JS-gated; DuckDuckGo/Startpage/SearXNG/Mojeeek CAPTCHA or 403; **Bing served deliberately poisoned results to scrapers** — it returned football and airline pages for "maximum adverse excursion stop loss"). Archive.org went offline partway through. Genuine content gaps that remain: **no audited intraday gold track record exists anywhere in the academic literature** (every gold performance number in this report is a backtest); **no peer-reviewed intraday gold-DXY beta**; **no post-2015 event study of the new**

LBMA auction's minute-by-minute impact on GC (a real gap we could fill with our own data); **no study of MGC specifically**; the canonical hidden-liquidity paper (Hautsch & Huang) and the VPIN literature are unread; and **Maximum Adverse Excursion is essentially absent from the academic literature** (zero relevant OpenAlex hits) — it lives in Sweeney's 1996/97 Wiley books and vendor software, and I reached no primary source. The method needs no citation, which is why it is item #1 on Monday. **Also under-covered by construction: YouTube, Twitter/X, and Discord** — which is where much of the "\$1,000/day" discourse actually lives. Anyone repeating this exercise should raise `CLAUDE_CODE_MAX_WEB_SEARCHES_PER_SESSION` first.

GAZBOT V7 · literature review 2026-09-13 · fifteen lenses, three syntheses, no backtests run, no service touched, nothing written outside the report tree. Full notes: `/home/alphabot/gazbot7/reports/lit_review_2026-09-13/` — 22,300 lines across 21 NOTES.md files, every claim carrying its URL and an A/B/C grade.